The Rise of H-2A: Causal Evidence on Guestworker Substitution for Settled Farm Labor *

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Abstract

Settled non-citizen workers, two-thirds of whom lack work authorization, have long supplied most seasonal farm labor in the US. In recent years, this workforce has declined during a period of stepped-up immigration enforcement. At the same time, the H-2A guestworker visa program, which allows farms to legally hire foreign workers, has rapidly expanded. This paper asks whether the decline in settled non-citizen farm workers contributed to the rise of the H-2A program, and at what rate. To address endogeneity, I instrument the contemporaneous settled non-citizen share of farm labor with its 2000 baseline share interacted with national enforcement shifts. Farms hire about one H-2A worker for each settled worker lost. The substitution accounts for 29 percent of aggregate H-2A growth since 2008. The effect appears only in crops, where the H-2A program is available, and not in year-round livestock. Farms are moving toward hiring through legal channels, and this shift has implications for agricultural and immigration policy.

JEL codes: J43, J61, J68, K37, Q12.

Keywords: H-2A program; guestworkers; immigration enforcement; farm labor; shift-share instruments; substitution.

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1 Introduction

The U.S. hired farm workforce can be divided into three parts: domestic workers, settled non-citizen workers, and seasonal guestworkers hired through the H-2A visa program. Domestic workers comprise U.S.-born and naturalized citizens, and their numbers have remained largely stable. Settled non-citizen workers, who have long been the largest source of seasonal farm labor, are workers who have established residence in the U.S., and two thirds of them lack work authorization. Their numbers have declined since 2015. Finally, H-2A guestworkers do the same seasonal farm work as settled non-citizen labor, with the difference that they are hired legally and for a short period, averaging 5.7 months. H-2A certifications grew from about 87,000 positions in 2008 to 385,000 in 2024, an increase of more than fourfold. Figure 1 depicts these trends: H-2A rises and settled non-citizen labor falls while the total remains stable. The two series measure different things and are not comparable in levels. H-2A positions are annual certifications, each covering under six months of work, while the American Community Survey (ACS) series count workers employed at the time of the survey.

This paper asks whether the decline in settled farm workers contributed to the rise of the H-2A program, and the answer is not straightforward. The two series can move in opposite directions without one replacing the other. The H-2A program can expand independently of the decline in settled non-citizen labor simply because the program becomes easier to use or because labor demand increases. Similarly, the settled non-citizen workforce can decline independently of the availability of H-2A workers, for example because fewer young workers migrate from rural Mexico. Separating substitution from these alternatives gives the rate at which H-2A guestworkers replace settled non-citizen workers, a number that matters for agricultural policy.

The substitution rate matters for three reasons. First, it informs policy aimed at ensuring adequate farm labor: whether the H-2A program can supply agricultural labor when the settled workforce contracts. This has been an ongoing topic of debate given enforcement pressures (Martin and Rutledge, 2026). Second, H-2A workers are generally more expensive than settled non-citizen labor (Simnitt and Onel, 2023), because employers must pay them a wage floor[1] and provide housing and transportation, which settled workers do not require. A substitution rate near one therefore implies that farms pay more than before for the same work.[2] Third, the rate of substitution shapes how farms adjust to the loss of settled workers.

[1] H-2A employers must pay at least the highest of the Adverse Effect Wage Rate (AEWR), the prevailing wage, and the federal or state minimum wage, and they must pay the same rate to U.S. workers doing the same work. A farm that does not use H-2A faces only the minimum wage.

[2] The wage floor has been rewritten and litigated twice since 2025, after the sample period ends (Congressional Research Service, 2025). When the Department of Labor issued the October 2025 rule, it projected that the lower wage floor would bring about 119,000 additional H-2A workers (U.S. Department of Labor,

Given the fixed costs of hiring through H-2A, small farms may find the program difficult to use (Castillo et al., 2024) and fall back on consolidation or mechanization (Clemens et al., 2018; Hamilton et al., 2022).

To find the substitution rate, I use a panel of 740 commuting zones from 2008 to 2024. The regressor of interest is the settled non-citizen share of agricultural laborers. I instrument that share with each zone's settled non-citizen share in 2000 interacted with two national enforcement series, the real Border Patrol budget and border encounters during the agricultural season. Commuting zones that had a higher share of settled non-citizen workers in 2000 are affected more by enforcement. The predicted gap between high- and low-exposure zones grows in years when national enforcement is higher.

I find that farms hire approximately one H-2A worker for each settled non-citizen worker they lose. This estimate is based on the Border Patrol budget instrument.[3] Estimates using the encounter and combined instruments are 1.6 and 1.2, respectively. Since H-2A contracts are typically shorter than the work year of settled non-citizen workers, H-2A replaces 0.7 of the lost labor in full-time equivalents. Substitution accounts for 29 percent of national H-2A growth between 2008 and 2024 with the budget instrument, and between 34 and 46 percent with the other two instruments. Event studies show no upward pre-trend in H-2A hiring in high-exposure zones; the early coefficients are negative. Settled workers in exposed zones began to decline faster than in less exposed zones in 2015, followed by a rise in H-2A hiring from 2019. This timing is consistent with substitution. The effect is observed only in crop production, where the H-2A program is available, and not in livestock. Other tests show that the effect is identified from settled non-citizen farm labor exposure, not from general immigrant density or proximity to the border.

Several papers study what drives H-2A growth. Castillo and Charlton (2023) argue that the housing boom of the 2000s pulled workers out of agriculture and into construction, which led to higher H-2A employment within commuting zones. Arteaga and Shenoy (2024) find that a decline in Mexican migration, measured by new arrivals rather than settled workers, raises wages and H-2A certifications. Castillo et al. (2023) attribute the growth to declining domestic farm employment, which they measure as U.S.-based Mexican-born farmworkers; their estimates show eighty new H-2A positions certified for every hundred such workers lost, over a sample ending in 2017. I measure the decline among settled non-citizens only, the group most affected by enforcement, since a majority of them lack work authorization. Over this paper's sample period, citizen farm employment rose rather than fell in the places

Employment and Training Administration, 2025).

[3]It gives the smallest estimates of both the substitution rate and the percentage of H-2A growth that can be explained by substitution; thus, the benchmark is the most conservative.

that lost settled workers, so the substitution documented here is between H-2A and settled non-citizen workers rather than between foreign and citizen workers. Extending the sample through 2024 also covers the years in which most of the program's growth occurred.

Earlier studies find that immigration enforcement has reduced the immigrant workforce on U.S. farms (Martin and Rutledge, 2026; Rutledge and Win, 2026). The effects of interior enforcement programs like 287(g) agreements and E-Verify mandates have been shown to reduce immigrant farm labor and raise farm costs (Kostandini et al., 2014; Charlton and Kostandini, 2021; Ifft and Jodlowski, 2022; Luo and Kostandini, 2022; East et al., 2023). Immigration arrests reduce non-citizen farm labor supply immediately (Gutiérrez-Li and Rubalcaba, 2025). The closest study in method is Amuedo-Dorantes et al. (2021), who find that interior enforcement raises firms' demand for H-2B non-agricultural guestworkers. They regress H-2B requests on a county index of local enforcement policies, whereas I instrument the settled farm workforce with national enforcement series interacted with baseline exposure, following the shift-share literature (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2025).

U.S. farms have experienced shortages of farm labor before. When the Bracero program ended in 1964, domestic workers did not replace the excluded Mexican workers; instead, farms mechanized or changed crops (Clemens et al., 2018). Even now, domestic applicants for farm jobs are nearly nonexistent (Clemens, 2022), and farms adjust on other margins, through mechanization, crop choice, and imports (Rutledge and Mérel, 2023; Hamilton et al., 2022; Charlton, 2025). This paper shows another way farms adjust, by hiring through legal channels like the H-2A program.

The rest of the paper proceeds as follows. Section 2 describes the program, the workforce, and the data. Section 3 sets out the specification and the instruments. Section 4 reports the main estimates, the event studies, the aggregate attribution, and inference for the substitution rate. Section 5 tests the identifying assumption one threat at a time. Section 6 concludes.

2 Background and Data

2.1 The H-2A program

The Immigration Reform and Control Act (IRCA) of 1986 created the H-2A visa program and also made it unlawful to knowingly hire unauthorized workers on farms. The program allows agricultural employers to hire foreign farm workers for temporary or seasonal jobs, meaning jobs that last less than a year. Year-round operations such as dairies do not qualify. To hire through this program, an employer must demonstrate a failed attempt to recruit domestic workers. Employers are also required to pay the Adverse Effect Wage Rate, a regionally set

wage floor based on average farm wages, provide housing that passes inspection, and cover transportation to and from the worker's home country.

The cost of using the program is high. Castillo et al. (2024) estimate that non-wage costs, primarily housing and transportation, are at least $10,000 for each H-2A worker. A portion of these costs is fixed per application, such as government filing fees and agent fees of $1,500 to $3,500 per application, so the cost per worker is higher for small applications. Employers must also file job orders 60 to 75 days before work begins, so hiring through the program requires about two months of lead time. The lead time delays the arrival of workers, and the high fixed cost discourages small farms that need only a few seasonal workers from using the program. As a result, the majority of H-2A use occurs on large farms and by farm labor contractors who recruit H-2A workers and provide them to multiple growers (Castillo et al., 2024; Rutledge et al., 2026). Two implications of this are important for the study.

First, since entry into the program is slow and costly, a farm that loses workers cannot quickly replace those workers through the program, especially if the farm has never used the program before.

Second, since the program does not allow for year-round employment, only farms with seasonal jobs can use the program to replace workers.

Workers do not qualify for the program on their own; instead, they must be sponsored by an employer. To sponsor workers, an employer must first obtain a temporary labor certification from the Department of Labor demonstrating that there are insufficient U.S. workers available and that hiring the workers will not negatively impact the wages of similarly employed U.S. workers. Once the certification is obtained, the employer may petition U.S. Citizenship and Immigration Services for the workers. After obtaining visas from their home countries, the workers may enter the United States. In general, petitions are only approvable for workers who are nationals of countries included on a list published annually by the Department of Homeland Security.[4] Workers must also meet the requirements of immigration law: unlawful presence in the United States for more than 180 days results in a three-year bar on re-entry; unlawful presence for more than one year results in a ten-year bar; and a previous removal order results in its own bar.[5] Therefore, settled workers removed through enforcement are generally ineligible to return through the program for years after being removed. The workers provided by H-2A are different people from the workers removed through enforcement.

Although adoption was very low during the first twenty years of its existence, it increased

[4]USCIS, "H-2A Temporary Agricultural Workers"; 8 CFR 214.2(h). Mexico has been on the eligible-country list in every year of the sample period.

[5]INA §212(a)(9).

significantly over the sample period as shown in Figure 1. Growth has occurred in every state; however, the largest growth has occurred in the fruit, vegetable, and nursery regions of California, the Pacific Northwest, the Southeast, and Florida.

2.2 The settled non-citizen workforce

After the end of the Bracero program in 1964, U.S. agriculture relied upon unauthorized migration from Mexico through migration networks and limited enforcement (Massey et al., 2016; Hanson, 2006). Many first migrated seasonally to the same region and often the same employer each year. As border enforcement increased the cost of crossing, immigrant workers settled in the United States and became the core of the seasonal workforce in labor-intensive crops (Massey et al., 2016). This workforce is now shrinking due to two factors. One factor is demographic: rural Mexico has been transitioning out of farm work for twenty years, and the number of new migrants has decreased accordingly (Fan et al., 2015; Hertz and Zahniser, 2013; Charlton et al., 2019). Another factor is policy: interior and border enforcement increased during the sample period, and a large body of literature establishes that enforcement reduces immigrant farm labor supply at the local level (Kostandini et al., 2014; Charlton and Kostandini, 2021; Ifft and Jodlowski, 2022; Luo and Kostandini, 2022; Gutiérrez-Li and Rubalcaba, 2025).

Legal status is not recorded in any available data set with geographic detail. The National Agricultural Workers Survey (NAWS) is the only U.S. survey that collects self-reported status but is not identified below the regional level. Therefore, I use settled non-citizen status in the ACS as the observable proxy for unauthorized status. In the fiscal 2021–2022 NAWS microdata, settled non-citizens make up 57.7 percent of hired crop workers, and 66.8 percent of them are unauthorized (U.S. Department of Labor, Employment and Training Administration, 2023). Approximately two-thirds of the group my measure captures lack work authorization. Since the settled non-citizen variable does not include workers who resided outside of the U.S. in the year prior to the ACS, H-2A visa holders are excluded. Appendix F contains the breakdown of the sample used for analysis and the diagnostic used to verify the exclusion. The ACS may undercount the settled workforce because unauthorized workers may be reluctant to participate in the survey. The effect of an undercount on the substitution rate depends on whether the undercount remains constant or increases as enforcement increases. If the ACS consistently undercounts the same proportion of settled workers each year, then the measured decrease in employment will be less than the true decrease in employment and the substitution rate will overstate replacement.

On the other hand, if workers become increasingly reluctant to participate in surveys as

enforcement increases, then some of the measured decrease in employment will result from workers failing to participate in surveys rather than leaving their jobs and the substitution rate will understate replacement. Under these circumstances, the instrument could lower the measured count without any worker leaving, and the design cannot separate the two.

2.3 Data and Sample

The panel includes 740 commuting zones (as defined by the 1990 delineation used by Autor et al. (2013)) from 2008 to 2024, except 2020 because the Census Bureau did not release a standard one-year American Community Survey (ACS) that year.[6] County and PUMA data are aggregated to commuting zones using Dorn (2009) crosswalks and extended to the 2020 PUMA delineation through Geocorr (Appendix A). Of the 11,840 commuting zone-year cells, 11,672 have a non-missing settled non-citizen share and make up the estimation sample.

The dependent variable is the number of H-2A positions certified per commuting zone and year from the Department of Labor's Office of Foreign Labor Certification disclosure files. Each record is an employer's application for H-2A workers; it contains the number of workers requested and certified, the requested employment dates, and the worksite location. I use case numbers to identify duplicate applications and then group those applications into commuting zones based on the county where the worksite is located. Certifications are the most commonly used measure in this area of research (Castillo and Charlton, 2023), since they are available annually, can be precisely identified geographically, and directly reflect employer demand.

The main regressor is the settled non-citizen share of agricultural laborers (in percent) from the ACS. An agricultural laborer is an employed individual aged 16 to 64 working in an agricultural industry and a hired farm-labor occupation; this excludes operators, managers, and clerical workers and matches the hired workforce covered by NAWS. A farm worker is considered a settled non-citizen if they are not a U.S. citizen and were not living abroad one year prior. I refer to this measure as PANCAL, for prior-arrival non-citizen agricultural labor.

Each zone's exposure is its settled non-citizen share of agricultural laborers in 2000 from the 5% sample of the decennial census with the same industry, occupation, and citizenship filters.[7] The national shifters are the log real Border Patrol budget and the log number of U.S.-Mexico border encounters between March and September, the months in which H-2A

[6]The Census Bureau released one-year ACS estimates for 2020 as experimental data due to low response rates caused by the COVID-19 pandemic. I exclude the year rather than combine the experimental file with the standard releases for consistency.

[7]The 2000 census does not ask where a person lived one year prior; I define settled workers based on years in the United States instead.

requested start dates tend to concentrate. The crop control is the vegetable share of cropland from the 2002 Census of Agriculture.

Table 1 presents summary statistics for the estimation sample and for commuting zones above and below the median 2000 exposure. High-exposure zones employ approximately four times as many H-2A workers as low-exposure zones and more than three times as many agricultural laborers. High-exposure zones also have larger Hispanic populations and vegetable cropland shares and are located closer to the U.S.-Mexico border. These differences motivate the crop controls and the tests of whether the effect is driven by farm labor rather than immigrant density or proximity to the border.

3 Empirical Strategy

This paper asks how many H-2A workers are hired, on average, for every PANCAL worker lost. The simplest estimator to answer this question in the absence of endogeneity is ordinary least squares (OLS). This section explains why OLS is expected to fail and sets out the strategy that identifies the causal effect.

3.1 Specification

The estimating equation is

$$\text{H2A}_{it} = \delta \, \text{PANCAL}_{it} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where i indexes commuting zones and t indexes years. H2A_{it} is the total number of H-2A certifications, and PANCAL_{it} is the percent of settled non-citizen agricultural workers. γ_i and λ_t are commuting zone and year fixed effects, respectively. $\mathbf{X}_{it}$ is the 2002 vegetable share of cropland interacted with year dummies (Appendix G). The regression is weighted by year-2000 agricultural employment, so that δ measures the effect per unit of farm labor, and standard errors are clustered by commuting zone.

The parameter of interest is δ, which captures the change in the number of H-2A workers hired for a one-percentage-point change in the PANCAL share. The sign is expected to be negative if farms substitute H-2A workers for settled non-citizen agricultural workers. For ease of interpretation, δ is converted into headcount and full-time-equivalent substitution rates.

3.2 Why ordinary least squares is expected to fail

I discuss three main sources of endogeneity in the OLS estimation of equation (1). First, there may be an omitted variable bias due to local labor demand shocks. While the crop controls included in $\mathbf{X}_{it}$ will capture the portion of labor demand that is related to the initial crop composition, labor demand will vary based on factors such as whether growers increase their total acreage, switch to more labor-intensive crops, open a packing facility near the area, or experience a drought. Therefore, the error term can be expressed as $\varepsilon_{it} = \theta D_{it} + \nu_{it}$, where D_{it} represents the local labor demand shock that is not captured by the controls. A positive demand shock both increases H-2A hiring ($\theta > 0$) and attracts or retains settled workers, which gives $\text{cov}(\text{PANCAL}_{it}, D_{it}) > 0$. The omitted-variable formula gives:

$$\text{plim}\,\hat{\delta}^{OLS} = \delta + \theta \frac{\text{cov}(\text{PANCAL}_{it}, D_{it})}{\text{var}(\text{PANCAL}_{it})} > \delta.$$

Thus, if $\delta < 0$, then the OLS estimate is biased toward zero and may even become positive.

Second, endogeneity may arise from measurement error in the independent variable. Classical measurement error causes attenuation of the coefficient toward zero. Most commuting zones have very small sample sizes in the ACS. Therefore, the year-to-year variation in the PANCAL share within a commuting zone can be due to sampling error.

Finally, reverse causality would push the OLS estimate away from zero. For instance, if the increase in H-2A hiring causes PANCAL workers to leave (e.g., because farmers now have access to legally available replacement workers), then the estimate will indicate substitution with the causal relationship reversed, resulting in the OLS estimate being pushed away from zero.

3.3 Instruments

I instrument the contemporaneous PANCAL share with two shift-share instruments by interacting a fixed measure of each commuting zone's baseline exposure with national enforcement series. Following Bartik (1991), the instruments are defined as follows:

$$(2) \qquad B_{it}^{BP} = s_{i,2000} \times \ln(\text{BP budget}_t), \qquad B_{it}^{enc} = s_{i,2000} \times \ln(\text{ag-season encounters}_t),$$

where $s_{i,2000}$ is the baseline PANCAL share of agricultural laborers in commuting zone i in year 2000. It is constructed from the decennial census, eight years before the panel begins. The two shift components are national series that measure enforcement intensity each year: the real Border Patrol budget and the number of agricultural-season border encounters.

An increase in national enforcement will affect high-exposure zones more than low-exposure zones and remove settled workers differentially, so the PANCAL share should fall by more in high-exposure zones. Since the instrument uses only baseline exposure and national conditions to predict this decrease, the zone's own demand for H-2A workers cannot affect it.

I use two shifters. First, the Border Patrol budget increases monotonically and represents the long-term growth of enforcement capacity. The second shifter, the agricultural-season encounter series, is non-monotonic and represents border activity during the months when farms hire. The instrument with the Border Patrol budget as the shift component is the benchmark throughout the paper because it gives the most conservative estimates. Although the budget and encounter instruments weight the years differently, they identify the same parameter if every commuting zone responds in the same way. The overidentification test, which uses both instruments, checks whether the two estimates are consistent with each other.

3.4 Identifying assumption

Identification is based on the baseline exposure shares, following the exogenous-shares framework developed by Goldsmith-Pinkham et al. (2020). While the national shifters give more weight to years with more enforcement pressure, they do not provide additional variation. Therefore, the identifying assumption is that, after accounting for fixed effects and crop controls, commuting zones with higher year-2000 exposure would have followed the same H-2A trajectory as less exposed zones if their settled share had not declined differentially. In other words, the instrument affects H-2A certifications only through the contemporaneous settled share.

Two features of the design support this assumption. First, the exposure share was determined prior to the H-2A expansion, the enforcement buildup, and the decline in the settled share, so it cannot be affected by them. Second, the exposure share is based on settled non-citizen agricultural labor rather than the overall immigrant population or domestic agricultural workers. This isolates the population most likely to be affected by enforcement and the population that can be replaced by the H-2A program.

3.5 First stage, reduced form, and event studies

The first stage regresses the PANCAL share on the instruments, with the same controls and fixed effects as in the main equation (1):

$$\text{PANCAL}_{it} = \pi^{BP} B_{it}^{BP} + \pi^{enc} B_{it}^{enc} + \mathbf{X}_{it}'\psi + \mu_i + \phi_t + u_{it}. \tag{3}$$

The reduced form regresses H-2A certifications on the instruments:

$$\text{H2A}_{it} = \rho^{BP} B_{it}^{BP} + \rho^{enc} B_{it}^{enc} + \mathbf{X}_{it}'\kappa + \alpha_i + \tau_t + e_{it}. \tag{4}$$

The just-identified models include only one instrument. The second stage of the IV model is equation (1) with the PANCAL share replaced by its predicted value from the first stage. For the just-identified models, $\hat{\delta} = \hat{\rho}/\hat{\pi}$. Table 2 presents results for all three stages, the Kleibergen-Paap F-statistic for instrument strength, and the Anderson-Rubin test, which is valid even if the first stage is weak. The Hansen J test checks whether the estimates from the two instruments are consistent with each other.

To examine the dynamic response, I replace the national shifter with a full set of year dummies:

$$Y_{it} = \sum_{k \neq 2008} \eta_k \left(s_{i,2000} \times \mathbf{1}[t = k] \right) + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}. \tag{5}$$

For the first-stage event study, Y_{it} is the PANCAL share. For the reduced-form event study, Y_{it} is H-2A certifications. With 2008 as the reference year, each η_k represents how the difference between more and less exposed commuting zones in year k compares to the same difference in 2008.

4 Results

4.1 Graphical evidence

Figure 2 plots two long differences against the 2000 PANCAL share. In panel (a), commuting zones with a higher baseline PANCAL share experienced larger declines in the contemporaneous PANCAL share between the 2008–2010 average and the 2022–2024 average; the slope is −0.18 (SE 0.04). Panel (a) is consistent with the relevance condition: places that historically had a larger share of settled workers are the ones where the contemporaneous PANCAL decline was largest.

Panel (b) shows that those same commuting zones also experienced larger increases in H-2A worker certifications over the same time period; the slope is 76.4 (SE 18.1).

Figure 3 illustrates how these two variables vary throughout the United States. Panel (a) shades commuting zones based on their baseline PANCAL exposure, whereas panel (b) shades commuting zones by the change in H-2A workers between the 2008–2010 and 2022–2024 averages. Most of the darkest areas in panel (b) correspond to the darkest baseline PANCAL exposure areas in panel (a); this is particularly true in California, Florida, and the Yakima

Valley/Columbia Basin region of Washington. Growth in the program has been concentrated in relatively few areas: thirty commuting zones make up approximately two-thirds of the total increase nationally.

4.2 Main estimates

Table 2, column (1), gives the OLS coefficient $\hat{\delta}^{OLS} = -2.11$ (SE 1.61), which is economically small and statistically insignificant. The OLS coefficient is subject both to classical measurement error from the thin-cell ACS share measure and to endogeneity.

Columns (2) to (4) report the two-stage least squares estimates. Instrumenting the PANCAL share with the Border Patrol budget instrument yields $\hat{\delta} = -156.7$ (61.3); the agricultural-season encounter instrument yields -226.5 (65.7); the overidentified specification, which uses both, yields -178.9 (60.7). The coefficient is interpreted as the change in the number of H-2A workers for a one-percentage-point change in the PANCAL share, so a negative sign indicates that a decrease in the settled share increases H-2A hiring.

For ease of interpretation I scale each coefficient by the workforce of the commuting zones that identify it. Two-stage least squares averages zone-level responses with weights that depend on the strength of the first stage in each zone, the Rotemberg weights of Goldsmith-Pinkham et al. (2020). Under the benchmark instrument, California alone carries about half of the weight because of its size and exposure. Zones whose shares move against the national pattern receive small negative weights. The weighted average workforce under the benchmark instrument is $\bar{N}^* = 15{,}426$ workers, so a one-percentage-point decline in the share represents about 154 workers. The estimates imply a substitution rate of 1.02 on the benchmark instrument and between 1.0 and 1.6 across the three specifications. H-2A contracts, however, are not year-round. The average contract lasts 5.7 months (Appendix Figure B1), while hired crop workers average roughly eight months of farm work per year.[8] Accounting for this difference, the full-time-equivalent substitution rate is 0.72 on the benchmark instrument and between 0.7 and 1.2 across specifications.

Panel B shows that the first stage is negative and precise for both instruments. The first stage has the expected sign: when a national shifter rises, the PANCAL share falls by more in commuting zones that had more settled workers to begin with. The Kleibergen-Paap (KP) F-statistics assess whether the instruments provide sufficient variation in the settled share for standard inference. They vary from 14.8 to 32.5 and exceed the 15 percent Stock-Yogo maximal size critical value. The Anderson-Rubin p-values, which are valid even when

[8] U.S. Department of Labor, National Agricultural Workers Survey, Public Data Tables, Table 2: hired crop workers averaged 35 weeks of farm work per year across the six survey cycles spanning 2007–2020 (35, 34, 35, 33, 35, and 39 weeks).

instruments are weak, reject a zero effect in all columns.

Panel C reports the reduced-form coefficients, which show that when a shifter rises, H-2A hiring rises by more in the more exposed commuting zones. In the just-identified columns, each reduced-form coefficient is equal to the product of the first stage and the second stage, up to rounding: $55.8 \approx (-0.356) \times (-156.7)$. When both instruments are included simultaneously, each instrument remains individually significant, indicating that each series provides additional predictive power beyond the other. The Hansen J test does not reject the overidentifying restriction ($p = 0.167$), indicating that there is no statistical difference between the two estimates.

4.3 The gap between OLS and IV

The IV estimates are approximately 75 to 105 times larger than the OLS estimates. I decompose this difference using an errors-in-variables (EIV) correction based on a direct measure of the reliability of the PANCAL share. I use a Monte Carlo simulation to obtain a reliability ratio by holding fixed each commuting zone's true PANCAL share across pooled years. For each commuting zone-year cell, I redraw the ACS sample using its actual respondent count, recalculate the share, and residualize it under the main specification. The estimated reliability is $\hat{R} = 0.180$.[9]

This implies that less than 20 percent of the within-zone variability in the measured share is "true" signal once fixed effects and other controls have been included. Classical measurement error biases OLS toward zero by this factor. Dividing the OLS estimate by $\hat{R}$ removes the bias and gives an EIV-corrected coefficient of -11.7. Since both the OLS coefficient and $\hat{R}$ are estimated, the correction carries uncertainty from both. A commuting-zone block bootstrap that re-estimates both in each replication yields a 95 percent confidence interval of $[-28.1,\ 8.2]$. Even the lower bound of this interval is far from the benchmark IV estimate of -156.7. Thus, measurement error alone cannot account for the gap.

The conclusion does not hinge on the estimated reliability: even if I assume a lower reliability of $R = 0.10$, the corrected OLS estimate would be only -21.1. Table 3 decomposes the gap between OLS and IV. Measurement error explains 6.2 percent of the difference between the OLS estimate (-2.1) and the benchmark IV estimate (-156.7); the remaining 93.8 percent is due to endogeneity. This is consistent with the omitted-variables explanation that positive local labor demand shocks increase H-2A hiring while also attracting or retaining settled workers. This creates a positive correlation between the error term and the PANCAL

[9]The reliability is one minus the ratio of the pure sampling variance to the total residual variance: $\hat{R} = 1 - 0.0140/0.0170 = 0.180$.

share, which biases OLS toward zero. Therefore, most of the difference between OLS and IV is due to endogeneity and not sampling noise.

4.4 Event study dynamics

The main estimates report a single coefficient for a pattern spanning sixteen years. Figure 4 shows how the relationship of baseline exposure with the settled share and with H-2A hiring evolves over time. I interact the year-2000 PANCAL share with a set of year dummies, with year 2008 as reference. Each coefficient captures how the gap in each variable evolves in high- versus low-exposure commuting zones relative to year 2008.

Panel (a) shows the first stage. The coefficients are not significant at the 5 percent level through 2014 and are not jointly different from zero over 2009–2012. They become significant at the 5 percent level from 2015. This means that the settled non-citizen share fell faster in exposed commuting zones than in less exposed ones from 2015 onward, and the gap widened in 2019 and remained wide through 2024. Panel (b) shows the reduced form. The early coefficients are small and negative, and jointly different from zero. The coefficients become positive and significant at the 5 percent level from 2019, rise through 2023, and remain large in 2024.

If H-2A hiring had been growing faster in commuting zones with a high baseline PANCAL share than in low-share zones before the settled share began to decline, the early coefficients in panel (b) would have been positive. Instead, the coefficients through 2012 are negative, which works against the effect rather than inflating it. The largest deviation is a dip in 2009, the recession year.

The two panels together depict the substitution pattern within a timeline. The settled share in exposed commuting zones first falls significantly in 2015, and H-2A hiring in the same zones first rises significantly in 2019. The gap between the two dates describes the timing; it is not an estimated lag, and Appendix C shows that the design cannot identify the lag length directly.

4.5 Aggregate attribution

Table 4 shows how much of the growth in H-2A employment can be attributed to the decline in the settled workforce. Between the 2008–2010 and 2022–2024 averages, H-2A employment increased by 298,923 workers, while the settled non-citizen agricultural workforce fell by 83,912. Both totals come from summing the changes across the 735 commuting zones.[10]

[10]The panel has 740 commuting zones. In five of them, the ACS samples no agricultural laborers in any year of one of the two windows, so the settled share is undefined and the change between windows cannot be

Multiplying the decline of 83,912 workers by the benchmark substitution rate gives a predicted increase of about 85,000 H-2A workers, which accounts for 29 percent of the observed growth. For the encounter and overidentified rates the shares are 46 and 34 percent, respectively. The settled-worker decline therefore accounts for between 29 and 46 percent of the program's growth. The remaining contributors include places that adopted H-2A without a history of reliance on settled non-citizen workers; Figure 3 shows moderate growth across the Midwest, where baseline exposure was low and H-2A use was minimal at baseline. They also include places where both the settled workforce and H-2A employment rose because local labor demand grew, variation the instrument deliberately sets aside.

4.6 Interpreting the magnitude

The main results in Table 2 report cluster-robust standard errors whose confidence intervals assume that $\hat{\delta}$ is approximately normal in large samples. With a sample of 740 commuting zones, this assumption would be reasonable, but when a large share of the weight is carried by a few commuting zones, the effective sample can be below 740. Therefore, I run a commuting-zone block bootstrap, which does not rely on this normality assumption. In each of the 999 replications, I draw the 740 commuting zones with replacement, re-estimate two-stage least squares, and form the substitution rate. Table 5 reports the resulting percentile intervals. The denominator is held at the full-sample first-stage average, $\bar{N}^* = 15{,}426$, so the rate's uncertainty comes from $\hat{\delta}$. Figure 6 illustrates how the sampling distribution is right-skewed; thus, the intervals are percentile-based as opposed to symmetric. Appendix E provides a version that also recalculates $\bar{N}^*$ in every replication. The headcount rate is approximately centered at one: the point estimate is 1.02, the bootstrap median is 1.11, and the 95 percent interval is $[0.28, 3.19]$. The rate is below one in 42 percent of the draws, so the data reject neither a rate below one nor a rate above one. The labor-time rate is centered below one, with a median of 0.79 and an interval of $[0.20, 2.26]$. For the headcount rate, a normal approximation using the 2SLS standard error yields an interval of $[0.24, 1.80]$. Although the lower limit of this interval is similar to the bootstrap interval, the upper limit understates the right tail, which reaches 3.19.

The right tail can be traced back to a single source as described in Appendix E. Farm labor is geographically concentrated; therefore, bootstrap draws that underweight the California core have less identifying variation, a weaker first stage, and larger $|\hat{\delta}|$. Conditioning on replications with a first-stage F above 10 trims the upper bound from 3.19 to 1.95 and increases the share of draws below one from 42 to 53 percent.

computed.

5 Robustness

The causal claim of the main results holds if the identifying assumption is valid: the instrument, baseline PANCAL exposure times national enforcement, affects the outcome variable, H-2A certifications, only through changes in the main regressor, the contemporaneous PANCAL share. There is no single way of testing this, so in this section I run a number of tests, each checking one way in which the estimate could be wrong. They also justify several choices made in constructing the variables and the specification: why the main regressor is the settled non-citizen share rather than a broader immigrant measure, how the estimate changes when the definition of the agricultural workforce is widened or narrowed, and why two instruments are used and how they differ.

5.1 Testing the exposure measure against immigrant density and border proximity

I test whether the substitution rate picks up the decline of settled non-citizen agricultural workers or simply captures higher immigrant density in general. The two are hard to tell apart. Suppose two commuting zones have the same baseline PANCAL share. The first has the majority of its immigrant workers in agriculture, and the second has a small fraction of a much larger immigrant population working in agriculture. When an enforcement shifter rises, farm workers in both zones are affected similarly, but in the second zone the larger immigrant workforce outside agriculture is also affected. Non-farm employers who lose immigrant workers may then draw other workers out of agriculture, such as citizens or recent arrivals not counted in the PANCAL share. If the second place also expanded H-2A hiring when the PANCAL share declined, because it lost these other farm workers rather than settled ones, and if places like that tend to have high PANCAL shares, the instrument would be picking up immigrant density in general and the coefficient would overstate substitution.

To test this, I build a non-agricultural version of the instrument in which the exposure share is the settled non-citizen share of the non-agricultural workforce, interacted with the same enforcement shifter. To make the non-agricultural instrument comparable to the agricultural one, I restrict the non-agricultural population to low-education workers. I then enter both instruments in one reduced form,

$$\text{H2A}_{it} = \rho^{ag} B_{it}^{ag} + \rho^{nonag} B_{it}^{nonag} + \mathbf{X}_{it}'\kappa + \alpha_i + \tau_t + e_{it}, \tag{6}$$

where B_{it}^{ag} is the agricultural instrument of the main specification and $B_{it}^{nonag} = s_{i,2000}^{nonag} \times \ln(\text{BP budget}_t)$ is its non-agricultural counterpart. If immigrant density were what mattered,

the two would split the effect or the non-agricultural instrument would dominate. Table 6 shows the opposite. The agricultural instrument keeps a positive, significant coefficient, while the non-agricultural instrument has the opposite sign and is imprecise. I also add the contemporaneous non-agricultural settled non-citizen share as a control in the main specification, and the estimate moves modestly, and away from zero, as Panel B of Table 6 shows.

Border enforcement is concentrated near the U.S.-Mexico border, and studies of its effects focus on border regions (Hanson et al., 2002; Bohn and Pugatch, 2015). Proximity to the border may therefore proxy for unauthorized activity and enforcement exposure. To see whether the variation actually comes from activity closer to the border, I interact negative log distance to the border with the same budget shifter as in the main specification. The second column of Table 6, Panel B, shows that this instrument is weak (first-stage F of 4.7) and gives an imprecise estimate. When put in a reduced-form horse race with the share-based instrument, the share-based instrument survives. Thus, proximity to the border does not predict H-2A hiring once the location of settled non-citizen workers is accounted for.

5.2 Testing for a response where H-2A is largely unavailable

H-2A is a seasonal program, limited to jobs that last less than a year. There are sectors that employ settled non-citizen workers but may not have access to H-2A because they require year-round employment. Livestock production is such a sector: dairy, its largest employer, along with cattle, hog, and poultry operations, relies on year-round labor, so enforcement may lead to a decline in settled workers but an H-2A alternative does not exist in most cases. If an H-2A response exists where the visa is not available, that would mean the results are picking up a different channel. If the response is substitution, it should be absent in livestock.

The main instrument's exposure is a broad definition of all agricultural laborers. Here, I split it into its two sectors. I define the crop share as the settled non-citizen share of laborers in crop production and landscape and horticultural services (IPUMS 1990 industries 10 and 20) and the livestock share as the settled non-citizen share of laborers in livestock production (industry 11), for each commuting zone and year.

The livestock test asks two questions. First, does enforcement remove settled livestock workers, creating a real shock for the placebo? The first stage answers this, with the contemporaneous livestock share as the dependent variable. Second, does that shock increase H-2A hiring? I answer this by entering both instruments in a single reduced form,

$$\text{H2A}_{it} = \rho^{crop} B_{it}^{crop} + \rho^{lvstk} B_{it}^{lvstk} + \mathbf{X}_{it}'\kappa + \alpha_i + \tau_t + e_{it}. \tag{7}$$

The 2000 values of each share are interacted with the enforcement shifter to give a crop instrument B_{it}^{crop} and a livestock instrument B_{it}^{lvstk}, constructed exactly as the main instrument. Table 7 reports both. The first stage for livestock is negative and significant with either enforcement series, implying that enforcement drains the livestock settled workforce as it does the crop workforce and the broader agricultural workforce in the main specification. However, in the horse race, H-2A certifications respond to the crop instrument while the coefficient on the livestock instrument is negative and imprecise. This means that the substitution response appears only where H-2A is available.

5.3 Testing whether H-2A in high-exposure zones was rising for other reasons

The substitution rate attributes the rise in H-2A hiring to the decline in settled non-citizen workers. However, it is possible that H-2A was replacing domestic labor, meaning U.S.-born and naturalized citizen workers, and the substitution rate picks that up. If domestic labor declined faster in high-exposure zones than in low-exposure zones in the same time frame when H-2A rose, then part of the H-2A growth could come from the decline in domestic labor.

To test this, I estimate the reduced form in equation (4) with domestic agricultural employment in place of H-2A certifications, entering one instrument at a time. Panel A of Table 8 shows that the coefficient is positive with both instruments, which means domestic employment rose in high-exposure zones relative to low-exposure zones as the shifters rose. H-2A workers in high-exposure zones were therefore replacing settled non-citizens, not citizens.

The event study for the domestic series in Figure 5 shows that domestic employment in high-exposure zones was already rising relative to low-exposure zones from 2009, and the joint test that the 2009–2012 coefficients are zero rejects at the 1 percent level.[11] This suggests that farm labor demand in high-exposure zones was growing from early in the sample. It is not a response to the settled decline: the settled share had not yet begun to fall in these years (the 2009–2012 first-stage coefficients are jointly zero, $p = 0.60$). The early H-2A coefficients are negative (Section 4.4), so this growth did not show up in H-2A hiring at first, but it could have in later periods, independent of the loss of settled workers.

Next, I control for this trend. Let G_i be commuting zone i's growth in domestic agricultural employment over 2008–2012, before the PANCAL decline began, so it is not itself a response to the decline. Its correlation with baseline exposure is only 0.18, so it does not absorb most

[11] The pre-trend holds for U.S.-born workers alone ($p = 0.004$), so it is not settled non-citizen workers naturalizing and being reclassified as citizens.

of the identifying variation. I add it to the main specification interacted with year dummies,

$$\text{H2A}_{it} = \delta \, \text{PANCAL}_{it} + \sum_{k \neq 2008} \omega_k \, G_i \times \mathbf{1}[t = k] + \mathbf{X}'_{it}\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{8}$$

so that each commuting zone follows its own time path scaled by its early domestic growth. Panel B of Table 8 shows that the estimate does not change qualitatively. The benchmark coefficient stays within half a standard error of its original value, although its first stage weakens ($F = 9.5$). The three estimates also converge: with the trend control they lie between -181 and -189, and the Hansen J p-value rises from 0.17 to 0.89. The substitution estimate does not rest on the domestic trend.

5.4 Testing whether the two instruments agree

This paper uses two instruments with the same baseline exposure share but different national enforcement shifters. The Border Patrol budget captures structural enforcement capacity, which grows monotonically year by year, while the agricultural-season encounter series captures contemporaneous border movement and is non-monotonic. Appendix Figure A1 plots the national series I use here. If responses to the loss of settled workers differ across zones, the 2SLS coefficient from each instrument is a weighted average of zone-level responses, and the weights differ across the two instruments (Imbens and Angrist, 1994; Angrist and Imbens, 1995). California, for example, carries 49 percent of the weight on the budget instrument and 37 percent of the weight on the encounter instrument. The two instruments could therefore give different estimates (Goldsmith-Pinkham et al., 2020). However, the Hansen J test ($p = 0.167$) shows that they are not statistically different even though, after fixed effects and controls, the correlation of their residual variation is only 0.28. This suggests that, even with different shifters, the two instruments agree on how the zones respond, which need not hold for every shifter.

Table 9, Panel A illustrates that other series that trend monotonically over the same period, such as the consumer price index, federal health spending, or a simple linear trend, produce coefficients similar to the Border Patrol budget's. Any trending series interacted with baseline exposure picks up the same variation across commuting zones. The encounter series, however, is different. When its values are randomly reordered across years, none of 500 placebo draws reproduces the first stage. Furthermore, the first-stage F-statistic of the encounter series is larger than that of 98.6 percent of random-walk shifters (Appendix Figure D1). The strength of the encounter instrument comes from the timing of its movements. Among placebo draws with a strong first stage, the coefficient is close to the benchmark.

Thus, the estimate is based upon the exposure share and not the specific shifter. At the same time, the estimate does not hold up with every random shifter. It works when the shifter moves over time together with the decline of the settled share, as both enforcement series do.

The immigration literature commonly uses a supply-shift instrument to allocate the national change in a specific group's population to those areas where that group was originally concentrated (Card, 2001). Following this literature, I construct an instrument with the national decline in PANCAL workers as the shifter and the baseline PANCAL share as the exposure, in a leave-one-out version and another version that includes the commuting zone's own workers. Panel B of Table 9 shows that it produces an estimate close to the enforcement instruments. I then test whether the supply-shift instrument agrees with the other instruments by pairing its leave-one-out version with the budget instrument, with the encounter instrument, and with both. The Hansen J test does not reject the overidentifying restriction at 5 percent in any pairing. An instrument whose shifter is the decline in settled workers from any cause gives an estimate that is not statistically different from the enforcement estimates. The agreement is closest with the encounter instrument, whose timing tracks the decline of the settled share most closely.

Table 10 reports the Rotemberg weights of Goldsmith-Pinkham et al. (2020) across the two instruments, across years, and across commuting zones. The overidentified estimate is a weighted average of the two just-identified estimates, with 68 percent of the weight assigned to the budget instrument and 32 percent to the encounter instrument. Across years, the weights on the budget and overidentified estimates are largest in 2019, 2023, and 2024. The years prior to 2018 have negative aggregate weight. On average, the shifters were below their panel means in those years and the settled share had not yet decreased, so those years serve as the comparison period for 2018–2024. Negative weights of this kind are a result of the decomposition, not a sign of misbehavior (Goldsmith-Pinkham et al., 2020). Across commuting zones, California carries 45 percent of the weight on the overidentified estimate. This reflects where U.S. farm labor is located and the weighting of the estimate by farm employment.

5.5 Alternative specifications and samples

Each row in Table 11 uses the overidentified specification, so that the Hansen J test can be reported. The comparison is therefore the overidentified estimate, -178.9, with a rate of 1.20, rather than the budget benchmark. The first group concerns California and weighting. Section 4.6 shows that the California core anchors the substitution rate, so the natural check is to remove it. Table 11 displays the results of dropping either all of California from the

sample or just its three largest zones. When I drop the three largest zones, leaving the rest of California in, the coefficient remains within a few units of the overidentified estimate, but the first stage becomes weaker ($F = 8.9$). Dropping all of California removes most of the identifying variation: the first stage weakens to an F of 5.5 and the coefficient, which doubles, is imprecise. No rate is reported for either row. The rate divides by the workforce of the zones that identify the estimate. That workforce drops from 14,913 workers to 1,065 when I drop all of California and to 3,953 when I drop the three largest zones, and the first stage weakens; thus, the denominator is no longer stable (Appendix E).

The unweighted model has similar implications. In this model, every commuting zone receives equal weight; however, a one-percentage-point decrease in PANCAL in California cannot be compared to a one-percentage-point decrease in Iowa. Therefore, the coefficient is smaller in levels, the identifying workforce is smaller, and the rate is 1.9, compared with 1.2 in the weighted specification. Both rates are between one and two. The weighted model measures the effect per unit of farm labor, which is what the substitution question asks. To determine whether the results are driven by a few extremely large users of H-2A, I drop the top 1 percent of commuting-zone-year H-2A outcomes. The coefficient remains close to the main estimate; however, the first stage becomes weak, because these observations carry much of the identifying variation.

The second group varies the sample and the timing. The crop agriculture sector covers crop production and landscape and horticultural services (IPUMS 1990 industries 10 and 20), and it is the sector that employs the most PANCAL and H-2A workers. Restricting the sample to crops gives a substitution rate close to two. The PANCAL construction specifically excludes recently arrived non-citizens. If recent arrivals were influencing my estimates, then adding their share as a control should move the main coefficient. However, as shown in Table 11, the coefficient barely moves, further supporting the evidence in Appendix F that recent arrivals do not contaminate the settled measure. Lagging the instruments by one year produces a nearly identical coefficient but a weaker first stage. Lagging the settled share instead of the instruments, which matches H-2A applications to the labor market conditions that existed one year earlier, produces a slightly larger coefficient. Neither of these specifications produces an estimate substantially different from the main estimate, and Appendix C explains why this research design cannot distinguish between these timings.

The next variation widens the definition of the main regressor to all Mexican-born agricultural laborers and to all foreign-born agricultural laborers. Both include citizens and recent arrivals. The broader definitions yield larger coefficients. These are the only two specifications in the paper where the Hansen J test rejects. The two enforcement instruments agree when the effect is measured on settled non-citizen agricultural labor but not on the

wider definitions, where the enforcement channel is diluted. The difference comes from the inclusion of citizens, who are unaffected by enforcement, and recent arrivals, who likely include H-2A workers themselves. Since the settled non-citizen share is the only definition for which both instruments give similar estimates, it serves as the paper's main regressor.

Splitting the panel into two halves, 2008–2017 and 2018–2024, I re-estimate the main specification in each. Neither half has enough identifying variation to estimate $\hat{\delta}$ precisely. The Rotemberg weights in Section 5.4 show why: the estimate compares the years after 2018 with the years before, and each half on its own loses that comparison. The point estimates follow the event-study timing, small in the early half and large in the late half.

Finally, Panel B of Table 11 shows the results when the outcome variable is proportional rather than in levels. When the outcome is the inverse hyperbolic sine of H-2A workers, the estimated coefficient is -0.089 (0.038). This indicates that a one-percentage-point reduction in the PANCAL share results in approximately a 9 percent increase in H-2A employment. When the outcome is the logarithm of H-2A workers, the 2,531 commuting-zone-years with zero H-2A are dropped (480 of them in commuting zones that never use the program). This gives a coefficient of -0.124 (0.037) with a first-stage F of 14.8, which means about a 13 percent increase in H-2A employment per percentage-point decline in PANCAL. The proportional transformations compress the large-H-2A zones that carry most of the identifying variation, so the implied rates, 0.59 and 0.90, are below the overidentified specification's 1.20.

6 Conclusion

This paper asks whether the decline in the number of settled non-citizen farmworkers contributed to the rise of the H-2A program, and if so, at what rate. For every settled worker lost, farms hire approximately one H-2A worker. Since H-2A contracts are typically short-term, this replaces approximately three quarters of the lost labor time. Replacement of lost settled workers accounts for approximately 29 percent of the total growth of the H-2A program between 2008 and 2024. Event studies show that places with a larger share of settled workers in 2000 experienced a faster decline in their settled share from 2015 and a faster increase in H-2A hiring from 2019.

This estimate is based on comparisons of commuting zones with a higher share of settled workers in 2000 versus those with a lower share. Therefore, it represents only one component of the program's expansion: H-2A hiring that replaced settled workers who departed. Many other factors contributed to the growth of the program. For example, use of the H-2A program increased throughout the Midwest, where very few settled workers worked in agriculture in 2000. Additionally, labor demand increased in certain areas, and the program may have

become easier to use. Some positions previously filled by new migrants from Mexico may now be filled through the H-2A program. None of these factors is included in the 29 percent. California carries about half of the identifying weight; thus, the rate reflects replacement primarily in areas where the settled workforce was largest.

The results have three implications. First, the H-2A program provides a way for farms to adjust their workforce after losing settled workers. Rather than simply doing less work, farms in these regions replaced most of the lost labor time. Second, replacing settled workers is costly. H-2A employers must pay a wage floor and provide housing and transportation, neither of which settled workers need. Therefore, the same work is more expensive, and changes to the wage floor and application fees directly affect the cost of adjusting. Finally, when farms can use the H-2A program, losing settled workers mainly changes who does the work instead of how much work is done. Settled workers living in the U.S. are replaced by temporary workers recruited from other countries. Since dairies and other year-round businesses cannot participate in the H-2A program, they do not have this option.

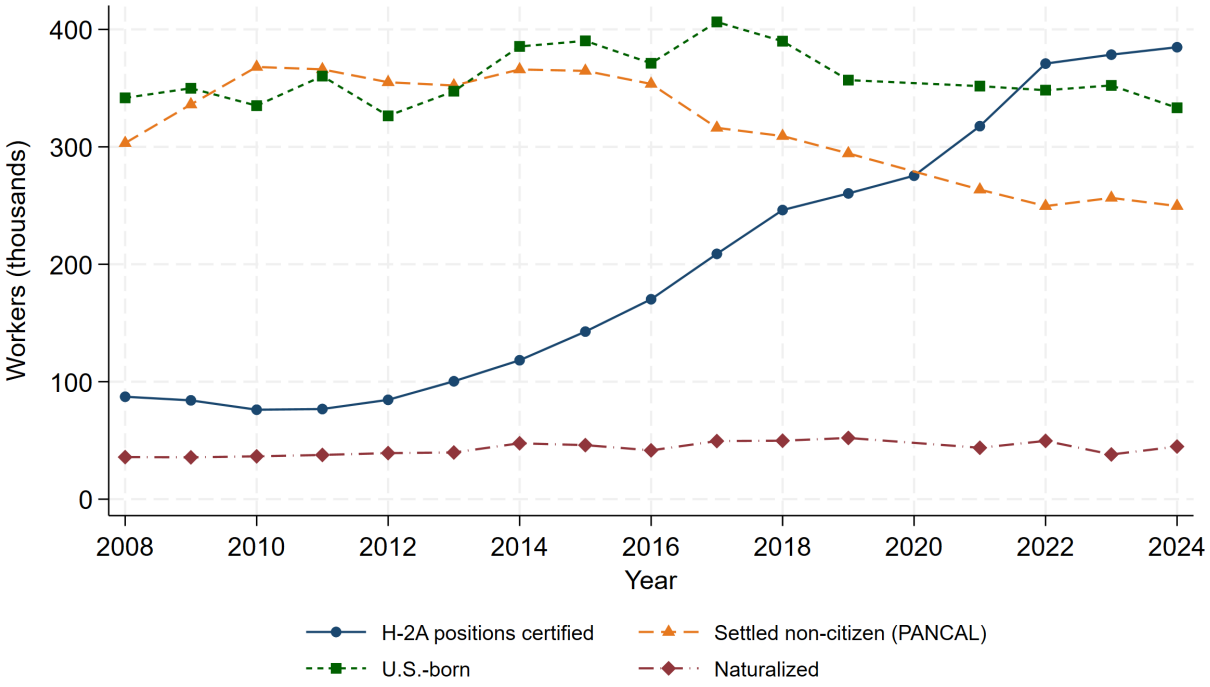


FIGURE 1. Components of the U.S. Hired Farm Workforce, 2008-2024

Notes: National totals in thousands of workers. H-2A positions certified are from the Department of Labor OFLC disclosure files, full and partial certifications, by fiscal year. The three ACS series are person-weighted counts of agricultural laborers, employed persons aged 16 to 64 in crop production, livestock production, or agricultural services in farm-worker, nursery-worker, or supervisor occupations, split by citizenship: settled non-citizen (PANCAL) workers are non-citizens who did not live abroad in the prior year, and the U.S.-born and naturalized series are citizens. The ACS did not release standard one-year estimates for 2020, so the ACS series have no value in that year. *Sources:* Department of Labor OFLC disclosure files; American Community Survey one-year files via IPUMS.

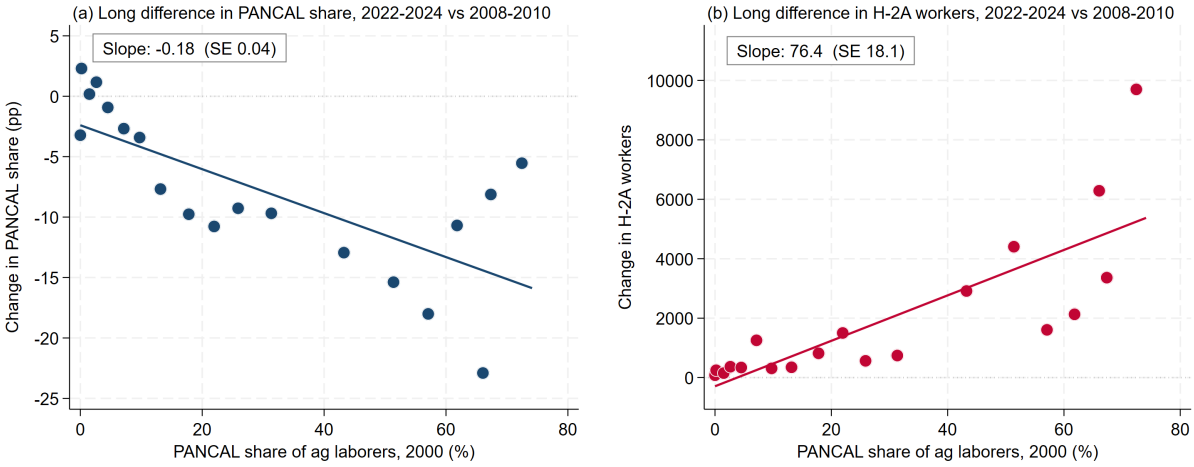


FIGURE 2. The Identifying Variation in Long Differences

Notes: Each panel plots a long difference against baseline exposure across the 740 commuting zones in the estimation sample. Baseline exposure is the PANCAL share of agricultural laborers in 2000. The long difference is the 2022–2024 average minus the 2008–2010 average, of the PANCAL share in percentage points in panel (a) and of certified H-2A workers in panel (b). Commuting zones are sorted by exposure and grouped into nineteen bins each holding approximately 5 percent of year-2000 agricultural employment; a commuting zone holding at least 5 percent by itself forms its own bin. Each dot is the employment-weighted mean of the bin. Fitted lines and the reported slopes are from regressions at the commuting-zone level, weighted by year-2000 agricultural employment, with heteroskedasticity-robust standard errors; they are not fits through the binned points. *Sources:* American Community Survey via IPUMS; Department of Labor OFLC disclosure files; Census 2000.

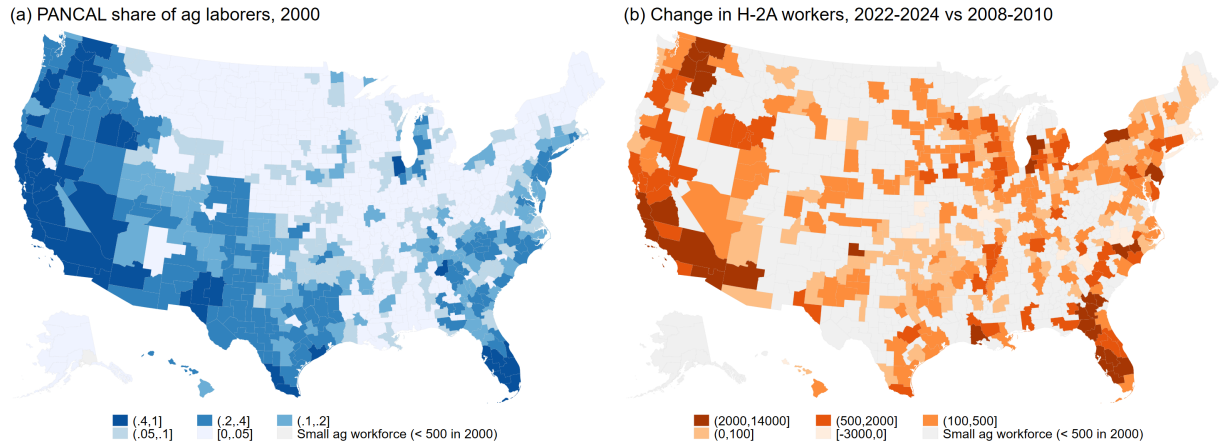


FIGURE 3. Baseline Exposure and the Growth of H-2A Employment, by Commuting Zone

Notes: Both panels show the 740 commuting zones in the estimation sample. Panel (a) shades zones by the PANCAL share of agricultural laborers in 2000, in round classes; the 10 percent boundary matches the high-exposure definition used elsewhere in the paper. Panel (b) shades zones by the change in certified H-2A workers, 2022–2024 average minus 2008–2010 average. Its class boundaries are zero, separating growth from no growth, and round values chosen so that the classes above zero hold 1.5, 11.8, 21.2, and 66.4 percent of the national increase, from lightest to darkest. Zones with fewer than 500 agricultural workers at baseline are left unshaded in both panels; the shaded zones carry 84.3 percent of year-2000 agricultural employment. *Sources:* American Community Survey via IPUMS; Census 2000; Department of Labor OFLC disclosure files; 1990 commuting zone boundaries.

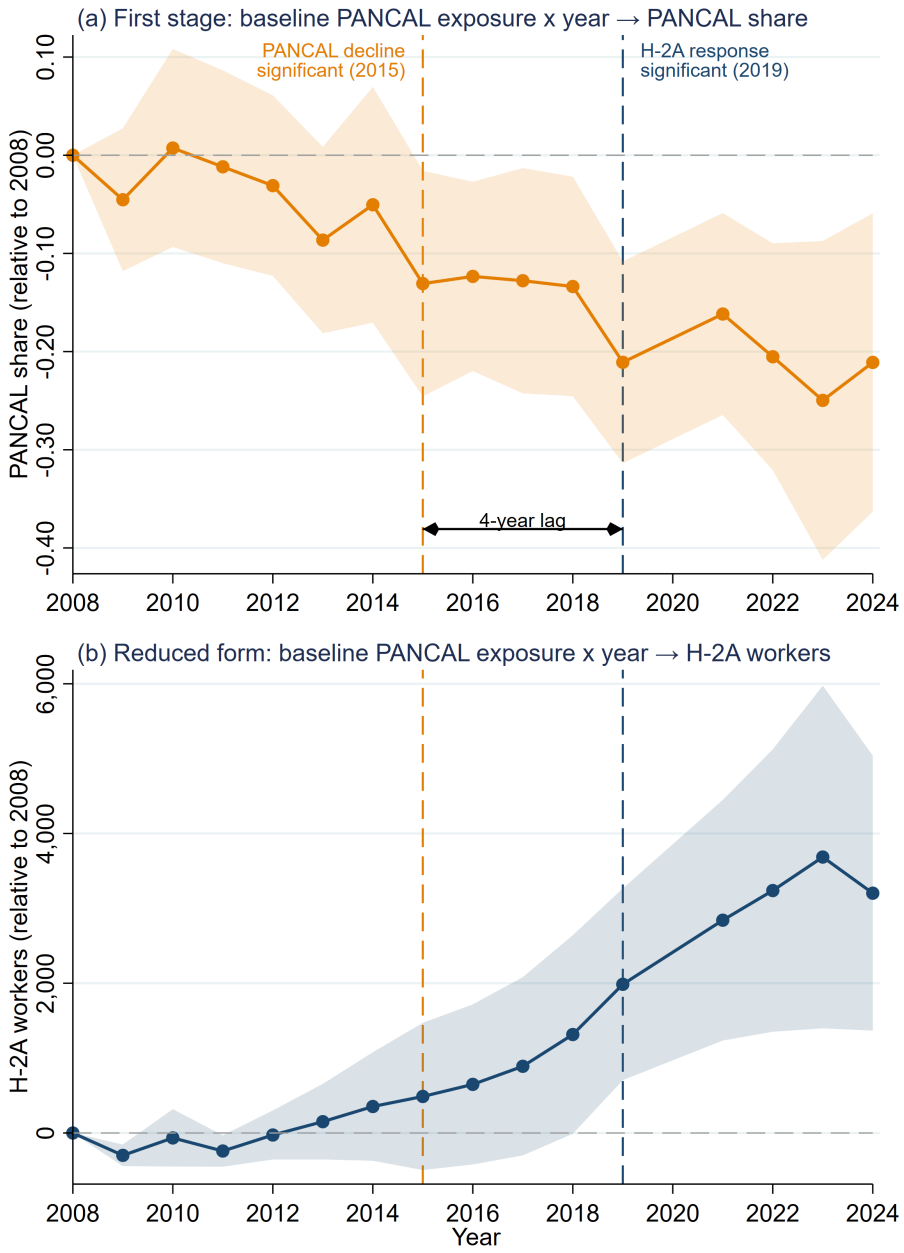


FIGURE 4. Event Studies of Baseline PANCAL Exposure

Notes: Coefficients from equation (5), with the PANCAL share as the outcome in panel (a) and H-2A workers in panel (b), estimated on the 740 commuting zones in the main sample with commuting zone and year fixed effects and vegetable share × year controls, weighted by year-2000 agricultural employment. There is no coefficient for 2020 because the ACS did not release standard one-year estimates for that year. Shaded bands are 95 percent confidence intervals clustered at the commuting zone level. Dashed vertical lines mark the first year in which the coefficient sequence is significant at the 5 percent level. *Sources:* American Community Survey via IPUMS; Department of Labor OFLC disclosure files.

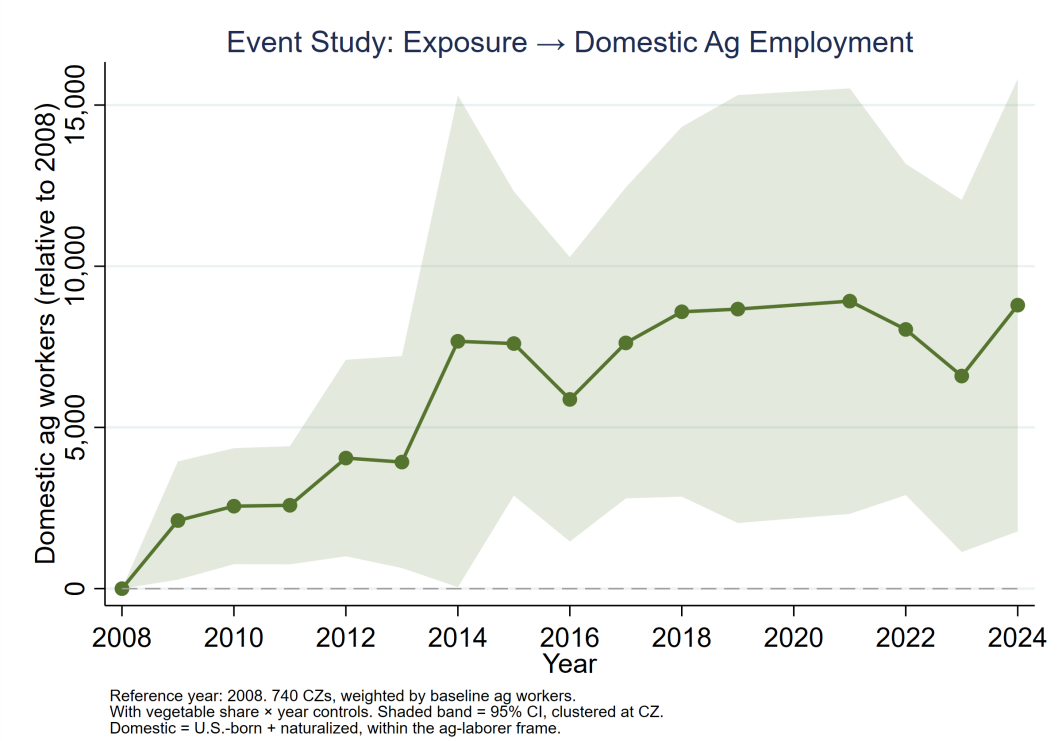


FIGURE 5. Event Study of Domestic Agricultural Employment on Baseline PANCAL Exposure

Notes: Each point is the coefficient on the year-2000 PANCAL share interacted with a year dummy, with domestic agricultural employment (U.S. citizens, born or naturalized) as the outcome and 2008 as the reference year. 740 commuting zones, weighted by year-2000 agricultural employment, with vegetable share × year controls. Shaded band is the 95 percent confidence interval clustered by commuting zone. The joint test that the 2009–2012 coefficients equal zero rejects at the 1 percent level.

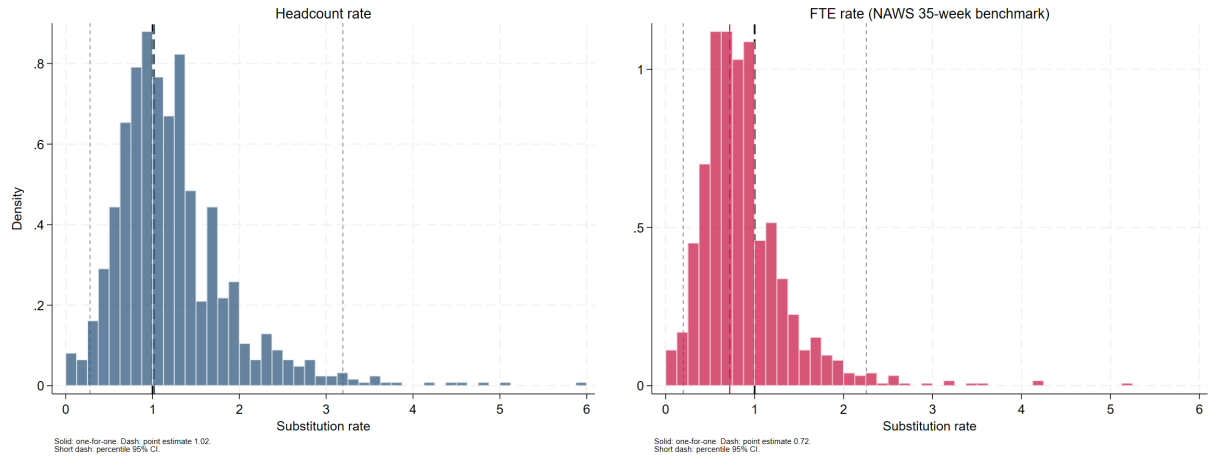


FIGURE 6. Bootstrap Distribution of the Substitution Rate

Notes: Histograms of the fixed-denominator substitution rate across 999 commuting-zone block bootstrap replications of Table 2, column (2), in headcount (left) and NAWS full-time-equivalent (right) units. The solid vertical line marks one-for-one replacement, the dashed line the full-sample point estimate, and the short-dashed lines the percentile 95 percent interval. Replications with a rate above 6 are not shown. See Table 5 and Appendix E.

TABLE 1. Summary Statistics

	(1) Full sample	(2) Low exposure (below median)	(3) High exposure (at or above median)
Panel A. Outcome and main regressor			
H-2A workers (count)	257.7	105.9	409.4
	(916.2)	(299.4)	(1,242.3)
PANCAL share (percent)	19.6	8.2	30.8
	(22.8)	(13.9)	(24.3)
Panel B. Agricultural employment and crop mix			
Total agricultural laborers	995.6	436.4	1,554.8
	(3,317.7)	(538.7)	(4,593.6)
Vegetable cropland share (percent)	1.0	0.3	1.8
	(3.4)	(0.9)	(4.6)
Panel C. Demographics and labor market			
Working-age population	280,794	128,718	432,870
	(779,586)	(281,307)	(1,044,134)
Hispanic share (percent)	11.4	4.5	18.3
	(14.7)	(3.7)	(18.0)
Employment rate (percent)	68.1	69.2	66.9
	(7.0)	(7.6)	(6.2)
Panel D. Geography			
Distance to U.S.-Mexico border (km)	1,503	1,726	1,281
	(763)	(716)	(744)
Panel E. Baseline (year-2000) shares, percent			
PANCAL share, 2000	12.0	1.4	22.7
	(14.7)	(1.8)	(14.2)
Non-agricultural settled non-citizen share, 2000	3.3	1.3	5.4
	(4.4)	(1.2)	(5.3)
Agricultural laborers, regression-weighted mean	9,062.5	—	—
Agricultural laborers, first-stage weighted mean ($\bar{N}^*$)	15,426.0	—	—
Observations (commuting zone-years)	11,840	5,920	5,920
Commuting zones	740	370	370

Notes: Panel means with standard deviations in parentheses; all shares and rates in percent. Column (1) is the full panel, 740 commuting zones over 2008 to 2024 excluding 2020; cells with zero agricultural employment have a missing PANCAL share and are dropped from estimation. Columns (2) and (3) split commuting zones at the median year-2000 PANCAL share of 6.0 percent. H-2A workers are certified positions, full and partial certifications. The full-sample mean of the 2000 PANCAL share, 12.0 percent, is below the pooled PANCAL mean in Panel A because the share rose between 2000 and the start of the panel: the unweighted mean is 21.2 percent in 2008–2010 and 17.0 percent in 2022–2024. Weighted by the 2000 agricultural workforce, the 2000 share is 30.1 percent and the panel share falls from 41.3 to 33.4 percent between the first and last three panel years; weighted means exceed unweighted means because farm labor is concentrated in high-share zones. The regression-weighted mean of agricultural laborers weights by the 2000 workforce; the first-stage weighted mean $\bar{N}^*$ is the denominator of the substitution rate (Section 4.2).

TABLE 2. The Effect of the Settled Non-Citizen Share on H-2A Employment

	(1) OLS	(2) Two-stage least squares	(3)	(4)
Excluded instruments	—	Budget	Encounters	Both
Panel A. Second stage. Dependent variable: H-2A workers				
PANCAL share (percent)	−2.11	−156.7	−226.5	−178.9
	(1.61)	(61.3)	(65.7)	(60.7)
Substitution rate (headcount)	—	1.02	1.64	1.20
Substitution rate (FTE, NAWS)	—	0.72	1.16	0.85
Panel B. First stage. Dependent variable: PANCAL share				
Border Patrol budget instrument	—	−0.356	—	−0.304
		(0.093)		(0.092)
Agricultural-season encounter instrument	—	—	−0.069	−0.047
			(0.012)	(0.010)
Kleibergen–Paap F-statistic	—	14.82	32.45	16.31
Panel C. Reduced form. Dependent variable: H-2A workers				
Border Patrol budget Bartik	—	55.8	—	41.9
		(16.1)		(13.1)
Agricultural-season encounter Bartik	—	—	15.5	12.6
			(4.0)	(3.3)
Anderson–Rubin p-value	—	< 0.001	< 0.001	< 0.001
Hansen J (p-value)	—	—	—	0.167
Observations	11,672	11,672	11,672	11,672

Notes: All regressions include commuting zone fixed effects, year fixed effects, and vegetable share × year interactions, and are weighted by year-2000 commuting zone agricultural employment. Standard errors, clustered at the commuting zone level (740 clusters), in parentheses. Panels B and C are estimated on the column (2)–(4) sample. In column (4), the reduced-form coefficients are from a single regression including both instruments; they are smaller than their single-instrument counterparts in columns (2) and (3) because the two instruments share the same exposure shares. The raw correlation between the two instruments is 0.997, but after fixed effects and controls the correlation of their residual variation is 0.28. The substitution rate is $|\hat{\delta}|/(0.01 \times \bar{N}^*)$, where $\bar{N}^*$ is the average zone agricultural workforce under the column's first-stage weights (15,426, 13,814, and 14,913 workers in columns (2) to (4)). Section 4.6 reports bootstrap intervals. *Sources:* Department of Labor OFLC disclosure files; American Community Survey via IPUMS; DHS Budget-in-Brief; CBP encounter records.

TABLE 3. Decomposing the OLS–IV Gap: Measurement Error versus Endogeneity

	(1) Coefficient	(2) Share of gap (percent)
Panel A. Estimates. Dependent variable: H-2A workers		
OLS	−2.11	—
	(1.61)	
Errors-in-variables-corrected OLS ($\hat{R} = 0.180$)	−11.7	—
	(9.4)	
Two-stage least squares (Border Patrol budget instrument)	−156.7	—
	(61.3)	
Panel B. Decomposition of the OLS–IV gap		
Total gap (IV minus OLS)	−154.6	100.0
Attributable to measurement error	−9.6	6.2
Attributable to endogeneity	−145.0	93.8
Panel C. Sensitivity of the corrected OLS to the reliability ratio		
Reliability $R = 0.10$	−21.1	12.3
Reliability $\hat{R} = 0.180$ (estimated)	−11.7	6.2
Reliability $R = 0.25$	−8.4	4.1
Reliability $R = 0.35$	−6.0	2.5

Notes: The reliability ratio R is the share of within-commuting-zone variance in the measured PANCAL share attributable to the true share rather than to sampling noise; across bootstrap replications $\hat{R}$ has mean 0.172 and standard deviation 0.019. The standard error of the corrected OLS estimate is from a block bootstrap with 999 replications resampling commuting zones, re-estimating both the coefficient and $\hat{R}$ in each. In Panel B the measurement error component is the corrected OLS estimate minus OLS; the endogeneity component is the IV estimate in Panel A minus the corrected OLS estimate. In Panel C, column (2) reports the share of the total gap that measurement error would explain at each value of R. OLS and IV specifications are as in Table 2.

TABLE 4. Share of H-2A Growth Explained by the PANCAL Decline

Instrument	Rate	Predicted ΔH-2A	Share explained (percent)
Border Patrol budget	1.02	85,243	28.5
Agricultural-season encounters	1.64	137,584	46.0
Both	1.20	100,683	33.7

Notes: The rate is $|\hat{\delta}|/(0.01 \times \bar{N}^*)$ from Table 2, columns (2)-(4). Predicted ΔH-2A is the rate times the decline in the settled non-citizen agricultural workforce, 83,912 workers, summed over the 735 commuting zones observed in both windows; the share divides by the observed H-2A growth of 298,923 over the same zones. Windows are the 2008–2010 and 2022–2024 averages.

TABLE 5. Bootstrap inference for the substitution rate

	Point	Bootstrap SE	Percentile 95 percent interval
Panel A. By unit			
Headcount	1.02	1.04	[0.28, 3.19]
FTE, NAWS 35-week	0.72	0.73	[0.20, 2.26]
FTE, 12-month	0.48	0.49	[0.13, 1.52]
Panel B. Headcount, by first-stage strength of the replication			
All replications (999)			[0.28, 3.19]
KP $F > 5$ (900)			[0.27, 2.30]
KP $F > 10$ (712)			[0.27, 1.95]

Notes: 999 commuting-zone block bootstrap replications of the benchmark specification. The rate divides $|\hat{\delta}|$ by $0.01 \times \bar{N}^*$, with $\bar{N}^*$ held at its full-sample value of 15,426, the average zone workforce under the first-stage weights. Appendix E reports the version that recomputes $\bar{N}^*$ in each replication. FTE conversions use the NAWS 35-week work year (factor 0.707) and a 12-month year (factor 0.476).

TABLE 6. Alternative Exposure Measures

Panel A. Horse races, reduced form. Dependent variable: H-2A workers

	Non-agricultural share	Border distance
Agricultural instrument (B^{ag})	74.2	74.1
	(26.3)	(23.3)
Alternative instrument	−82.3	−8.3
	(84.1)	(5.5)

Panel B. Two-stage least squares. Dependent variable: H-2A workers

	Non-agricultural share as control	Border distance as instrument
PANCAL share (percent)	−187.0	−28.0
	(81.6)	(59.8)
Kleibergen-Paap F	12.65	4.67

Notes: The agricultural instrument is the main specification's instrument, the year-2000 PANCAL share interacted with the Border Patrol budget. The alternative instrument in Panel A, column (1), interacts the year-2000 settled non-citizen share of low-education non-agricultural employment (high school or less, aged 16 to 64) with the same shifter; the two baseline shares have a correlation of 0.70 across commuting zones. In column (2) it interacts negative log distance to the Mexican border with the same shifter. Each column of Panel A enters the agricultural instrument and the alternative instrument in one reduced form, equation (6). Panel B, column (1) adds the contemporaneous non-agricultural settled share as a control to Table 2, column (2), which gives −156.7 (61.3) without it; column (2) instruments the PANCAL share with the distance instrument alone. Specification, weights, and clustering as in Table 2. Standard errors clustered by commuting zone in parentheses.

TABLE 7. The Livestock Placebo

	Border Patrol budget	Encounters
Panel A. First stage. Dependent variable: livestock PANCAL share (percent)		
Livestock instrument (B^{lvstk})	−0.57	−0.15
	(0.20)	(0.04)
Panel B. Horse race, equation (7). Dependent variable: H-2A workers		
Crop instrument (B^{crop})	92.2	
	(29.4)	
Livestock instrument (B^{lvstk})	−60.6	
	(34.2)	

Notes: Livestock is IPUMS 1990 industry code 11 (livestock production, including dairy, cattle, hogs, and poultry), with the same occupation and settled non-citizen filters as PANCAL. The crop and livestock instruments interact the year-2000 settled non-citizen shares of crop and livestock employment with the enforcement shifter; Panel B uses the Border Patrol budget shifter. Livestock cells in the ACS are thin, so the first stage in Panel A is read as a sign and significance check rather than a precise magnitude. Specification, weights, and clustering as in Table 2. Standard errors clustered by commuting zone in parentheses.

TABLE 8. Domestic Agricultural Employment and the Substitution Estimate

	Border Patrol budget	Encounters	Both
Panel A. Reduced form, domestic employment as the outcome			
Instrument	135.6	12.4	—
	(48.6)	(6.3)	
Panel B. H-2A workers as the outcome, with pre-period domestic growth × year controls			
PANCAL share (percent)	−181.0	−189.1	−184.8
	(97.4)	(72.5)	(81.9)
Kleibergen-Paap F	9.50	27.53	14.04
Hansen J (p-value)	—	—	0.89
Same specification, no control	−156.7	−226.5	−178.9

Notes: Domestic employment is agricultural laborers who are U.S. citizens, born or naturalized. Panel A estimates the reduced form in equation (4) with domestic employment in place of H-2A workers as the outcome, entering one instrument at a time; the coefficients of the two instruments are not comparable in size because the shifters are in different units. Panel B adds to the main specification each commuting zone's 2008–2012 domestic employment growth interacted with year dummies. Specification, weights, and clustering as in Table 2. Standard errors clustered by commuting zone in parentheses.

TABLE 9. Alternative Instruments

	$\hat{\delta}$	Kleibergen-Paap F	Hansen J (p-value)
Panel A. Smooth national shifters in place of the budget			
Consumer price index	−182.8 (63.7)	19.00	—
Federal health spending	−175.4 (59.0)	22.35	—
Linear trend	−168.7 (57.5)	22.36	—
Agricultural-season encounters (non-smooth)	−226.5 (65.7)	32.45	—
Panel B. Supply-shift instruments (baseline share × national PANCAL decline)			
Leave-one-out national share, alone	−222.4 (83.9)	16.10	—
National share, alone	−179.9 (60.5)	24.43	—
Leave-one-out share + budget	−177.2 (67.3)	8.40	0.08
Leave-one-out share + encounters	−223.9 (75.5)	17.18	0.92
All three instruments	−180.5 (61.6)	11.93	0.21

Notes: Each row is the specification of Table 2 with the instrument set named. In Panel A the year-2000 PANCAL share is interacted with the log of the named national series in place of the Border Patrol budget. In Panel B the supply-shift instrument interacts the year-2000 PANCAL share with the log of the national PANCAL share, computed excluding the commuting zone's own workers (leave-one-out) or including them. Standard errors clustered by commuting zone in parentheses.

TABLE 10. Rotemberg Weights: Where the Identification Comes From

	Budget	Encounters	Both
Weight in the overidentified estimate	0.68	0.32	—
Weights by period (sum of year weights)			
2008–2017	−23.9	−12.0	−20.1
2018–2024	24.9	13.0	21.1
Three largest year weights, largest first	2023, 2024, 2019	2023, 2024, 2022	2023, 2024, 2019
Weights by commuting zone			
California share of identifying weight	0.49	0.37	0.45
Zones with negative weight (of 740)	217	267	227

Notes: Rotemberg decomposition following Goldsmith-Pinkham et al. (2020). The instrument weights are the shares of the overidentified estimate attributable to each just-identified estimate; the weighted just-identified estimates reproduce −156.7, −226.5, and −178.9 exactly. Year weights and zone weights each sum to one within a column, and individual weights can be negative. The negative sum over 2008–2017 reflects shifters that were, on average, below their panel means while the settled share had not yet fallen; these years serve as the comparison period for 2018–2024. The California share is the sum of weights on California commuting zones.

TABLE 11. Specification and Sample Robustness

	$\hat{\delta}$		KP F	J (p)	Rate
Panel A. Levels, both instruments					
Main specification	−178.9	(60.7)	16.31	0.17	1.20
California and weighting					
Excluding California	−346.2	(162.8)	5.55	0.57	—
Excluding three largest California zones	−182.7	(103.5)	8.94	0.93	—
Unweighted	−65.9	(21.7)	28.99	0.52	1.94
Dropping top 1% of H-2A outcomes	−168.2	(94.7)	4.12	0.23	—
Sample and timing					
Crop agriculture only	−211.6	(78.3)	13.06	0.39	2.01
Recent-arrival share control (migration)	−172.9	(56.5)	16.58	0.08	1.06
Recent-arrival share control (years in U.S.)	−181.3	(62.4)	15.34	0.30	1.29
Instruments lagged one year	−179.9	(60.7)	12.02	—	1.41
PANCAL share lagged one year	−193.1	(65.0)	15.35	—	1.29
Wider definitions of the main regressor					
Mexican-born share	−214.3	(77.1)	9.47	0.04	—
Foreign-born share	−216.2	(75.3)	8.55	0.04	—
Panel halves					
2008–2017 only	−69.1	(59.2)	2.43	0.73	—
2018–2024 only	−325.4	(204.3)	1.90	—	—
Panel B. Proportional outcomes, both instruments					
asinh(H-2A workers)	−0.089	(0.038)	16.31	0.33	0.59
ln(H-2A workers), zeros dropped	−0.124	(0.037)	14.79	0.18	0.90

Notes: Each row is the overidentified specification of Table 2, column (4), with the change named. Standard errors clustered by commuting zone in parentheses. Rates are $|\hat{\delta}|/(0.01 \times \bar{N}^*)$ with $\bar{N}^*$ the average zone employment under the first-stage weights of the row's own specification; rates are not reported for rows with a Kleibergen-Paap F below 10, where the first-stage weights that define $\bar{N}^*$ are unstable (Appendix E). In Panel B the implied rate evaluates the proportional effect at the weighted sample-mean H-2A level (987 in the full sample; 1,080 among positive cells) and divides by the specification's $\bar{N}^*$; it is an accounting conversion, not a second estimate. The log specification drops 2,531 of 11,840 commuting-zone-years with zero H-2A workers, 480 of them in commuting zones that never use the program.

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A Data Construction

A.1 H-2A certified positions

The disclosure files include data from fiscal years 2008 through 2024. Because applications are deduplicated by case number, each multi-worksite filing has its certified position(s) counted only once. Worksites were assigned to counties based upon either the county field provided in the files (fiscal year 2017 forward), ZIP codes (fiscal years 2015 and 2016), or city name with manual assignment (fiscal years 2008 through 2014). Once counties were identified, they were mapped to 1990 commuting zones using the crosswalks developed by Dorn (2009) and Autor and Dorn (2013). Geocoding was successful for approximately 97 to 99.6 percent of positions in the county-field years and approximately 93 to 97.5 percent of positions in the city-name years. Therefore, a greater percentage of early-year positions is not assigned to a commuting zone.

An application may be certified in full or in part. When an application is partially certified, the number of workers approved is less than what the employer requested, and the certified count on these records represents the reduced amount. Both types of certifications are included in the final outcome, consistent with OFLC's reporting of total certified positions. The constructed series is within 3 percent of the reported total in all years except fiscal year 2008, when it is 6 percent greater than the reported figure (approximately 5,100 positions). This difference is approximately equal to the number of positions on partial certifications in the fiscal year 2008 file, indicating that the reported total for fiscal year 2008 likely did not account for partial determinations while the totals for subsequent years did.

Additionally, while OFLC's reported figures include Puerto Rico, the commuting-zone panel does not.

A.2 The settled non-citizen share

Agricultural laborers are identified with 1990 industry codes 10, 11, 20, and 30 (crop production, livestock production, landscape and horticultural services, and agricultural services) and 1990 occupation codes 479, 484, and 488 (farm workers, nursery workers, and supervisors). The share is computed with person weights. PUMAs are mapped to commuting zones with the same crosswalks for the 2000 and 2010 PUMA delineations; for the 2020 delineation, I build the crosswalk through Geocorr 2022.

For the exposure share $s_{i,2000}$, I define settled workers in the 2000 census as non-citizens with more than one year in the United States.

A.3 National enforcement series

The budget shifter is the annual Border Patrol budget in 2024 dollars, assembled from Department of Homeland Security Budget-in-Brief documents, Congressional Research Service report RL32562, and appropriations records, and deflated by the CPI-U. The encounter shifter is the number of U.S.-Mexico border encounters that Customs and Border Protection recorded from March through September. The comparison series in Section 5.4 are the CPI-U and real federal health spending, from the Bureau of Labor Statistics and the Bureau of Economic Analysis. Figure A1 plots the series.

A.4 Crop controls

Crop composition data were obtained from the 2002 Census of Agriculture via USDA Quick Stats at the county level; specifically, total cropland and acreage in vegetables, orchards, berries, and tobacco. Data were then aggregated into commuting zones using the same crosswalks, and each category was converted into a percentage of cropland.

TABLE A1. Variable Definitions and Sources

Variable	Years	Definition	Source
Outcome			
H-2A workers	2008-2024	Certified H-2A positions per commuting zone-year, full and partial certifications	DOL OFLC disclosure files
Main regressor			
PANCAL share	2008-2024	Non-citizen agricultural laborers who did not live abroad in the prior year, as a share of all agricultural laborers (percent)	ACS one-year via IPUMS
Instrument: exposure share			
PANCAL share, 2000	2000	Same construction, with years in the U.S. greater than one replacing the migration filter	2000 census 5 percent sample via IPUMS
Instrument: national shifters			
Border Patrol budget	2008-2024	Log real annual budget, 2024 dollars	DHS Budget-in-Brief; CRS RL32562; CPI-U
Ag-season encounters	2008-2024	Log March-September U.S.-Mexico border encounters	CBP encounter records
Controls			
Crop shares	2002	Vegetable, orchard, berry, and tobacco acreage as shares of cropland	USDA Census of Agriculture
Weights			
Agricultural employment, 2000	2000	Agricultural laborers per commuting zone	2000 census via IPUMS

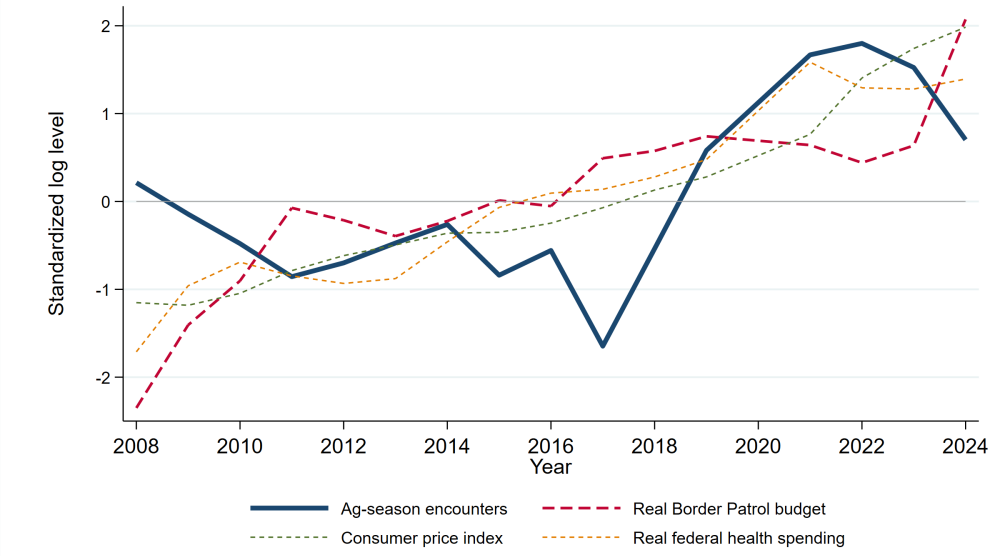


FIGURE A1. National Shifter Series Used to Construct the Instruments

Notes: Each series is the log of the national series, standardized to mean zero and standard deviation one over 2008 to 2024. A linear trend, also used in Section 5.4, is not plotted. *Sources:* DHS Budget-in-Brief; CBP encounter records; Bureau of Labor Statistics; Bureau of Economic Analysis.

B H-2A Contract Duration and the Labor-Time Conversion

The full-time-equivalent substitution rates use the average length of an H-2A contract to calculate the amount of labor time represented by headcount. Figure B1 shows the worker-weighted mean contract duration by year based upon the certification data. Duration was calculated as the requested employment end date minus the start date on each application. Contracts were approximately 6.4 to 6.8 months long through 2011 and have been approximately 5.5 months long since 2013, which covers the period in which the settled workforce declined and H-2A employment increased. Over the 2008 to 2024 period, the worker-weighted mean was 5.71 months. According to the NAWS, all hired crop workers averaged 35 weeks (or 8.08 months) of farm work per year during the six survey cycles that spanned 2007–2020. Therefore, there are two conversion factors: $5.71/8.08 = 0.707$ (NAWS benchmark) and $5.71/12 = 0.476$ (twelve-month benchmark).

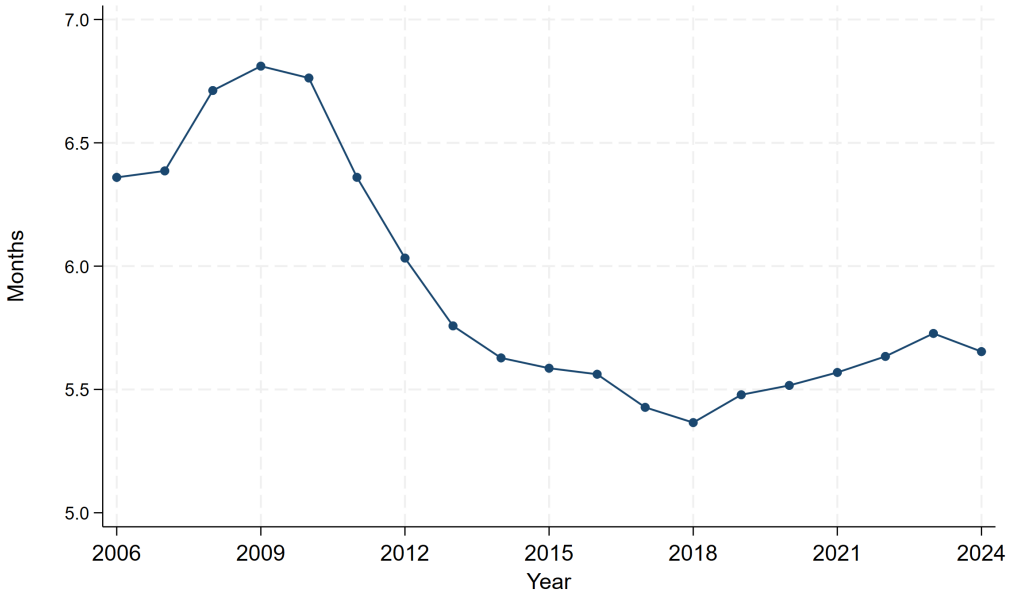


FIGURE B1. Average H-2A Contract Duration by Year

Notes: Worker-weighted mean length of certified H-2A contracts, in months, by fiscal year of certification. Contract length is the requested end date minus the requested start date on each employer application, weighted by the number of workers certified. Applications with missing, non-positive, or implausibly long durations (over 37 months) are dropped. *Sources:* Department of Labor OFLC H-2A disclosure files, 2006–2024.

C Lagged Instruments

Using lagged instruments, I test whether the four-year difference between the decline in the settled share (2015) and the increase in H-2A hiring (2019) is due to a delayed response by H-2A.

I run a regression of H-2A workers in year t on the PANCAL share in year $t-k$, instrumented with the budget instrument in year $t-k$, for $k = 1$ to 4. If H-2A responds with a delay, then the lagged share will predict H-2A hiring differently than the current share. Lagging drops the first k years, so each row also presents the contemporaneous estimate on the same observations; thus, the only difference between the two columns is timing. Lags that occur in 2020 are omitted since the ACS did not release any standard estimates for that year. All specifications, weights, and clustering are identical to those used in Table 2.

Table C1 displays the results. At all lags, the lagged and contemporaneous estimates are statistically indistinguishable. Both increase as k increases because the shorter samples are shifted toward later years where the effect is larger. A distributed-lag model that includes the share at time t and at time $t-4$ together cannot tell the two apart: the joint first-stage F statistic is 0.3, and neither coefficient is statistically significant on its own. Only their sum is statistically significant (-191.1, $p = 0.004$).

The reason one cannot tell apart the effects of the two shares is the shifter. The log budget has been growing smoothly over time. The budget in year $t-k$ is approximately equal to the budget in year t minus a roughly constant amount. When the log budget is multiplied by the 2000 share, the lagged instrument will be approximately equal to the current instrument minus a zone-specific constant. This zone-specific constant will be removed through the use of commuting zone fixed effects. As a result, both the lagged and current instruments carry almost the same variation. Thus, the design cannot identify the length of the lag. This is a general property of shifters that rise smoothly over time, which cannot separate the response to the current shock from ongoing adjustment to earlier ones (Jaeger et al., 2018).

TABLE C1. Lagged Instruments, Budget Instrument

Lag k	$\hat{\delta}$ at $t-k$	KP F	$\hat{\delta}$ at t, same sample	KP F	Observations
1	−174.1 (70.1)	13.7	−172.3 (67.5)	15.0	10,108
2	−180.2 (69.6)	16.5	−169.4 (66.2)	15.8	9,406
3	−200.3 (76.4)	13.4	−199.7 (85.9)	11.7	8,706
4	−215.9 (87.7)	10.9	−191.1 (69.7)	16.0	7,962

Notes: Each row instruments the PANCAL share at $t-k$ with the budget instrument at $t-k$ (columns 2–3) and, on the same observations, the share at t with the instrument at t (columns 4–5). The full-sample contemporaneous estimate is −156.7 (61.3). Standard errors clustered by commuting zone in parentheses. Specification and weights as in Table 2.

D Placebo Shifters

To test if the timing of the encounter series is what makes the encounter instrument strong, I use placebo series instead of the national shifter to run the benchmark model 500 times for each of three types: the encounter series with its years randomly reordered; the budget series with its years randomly reordered; and a random walk. When I reorder either series, most of the first stage disappears. The median F is 3.1 for reordered encounters and 3.3 for reordered budgets. None of the 500 reordered encounter series reaches the F of the actual series (32.5).

Random walks behave differently. The median F is 17.2, and 59 percent of the random walks are above the budget instrument's F of 14.8. A random walk trends, and any trending series interacted with baseline exposure picks up the faster decline of the settled share in high-exposure zones, the same pattern as Panel A of Table 9. Only 1.4 percent of random walks reach the encounter instrument's F.

The median coefficient for all three types of placebo draws (with a first-stage F greater than 10) is between -148 and -166; this is very close to the benchmark value of -156.7. The coefficient is based on the exposure share, and the timing of the encounter series is what distinguishes it from a trend.

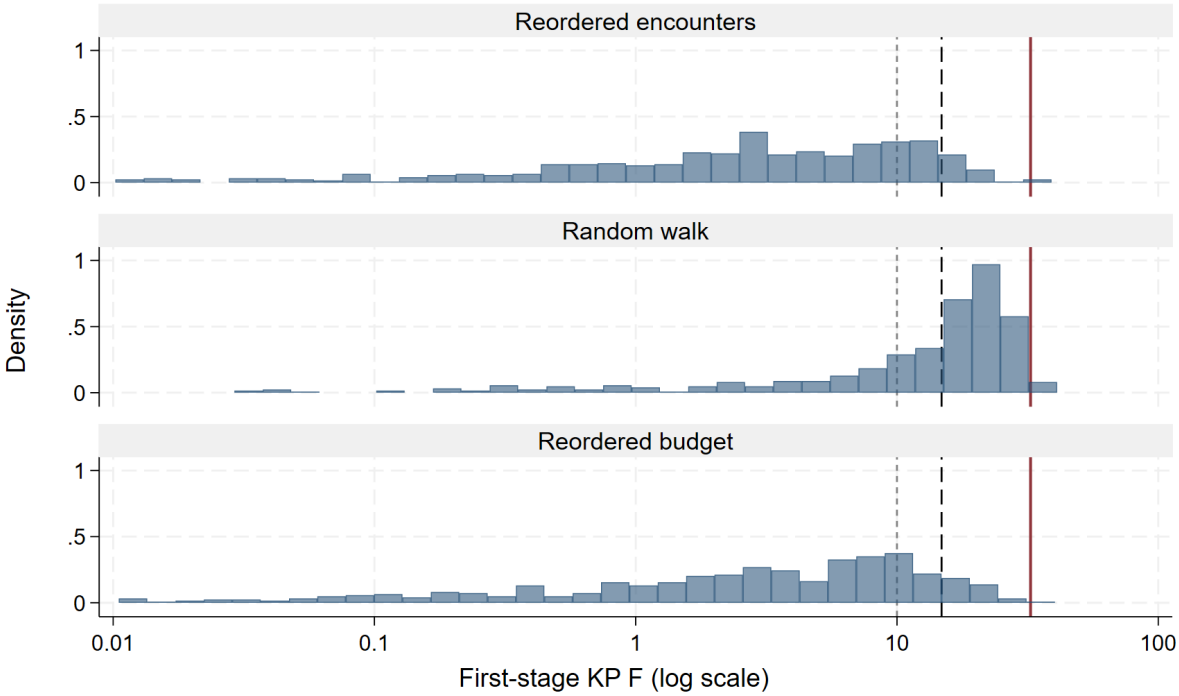


FIGURE D1. Placebo Shifters: First-Stage Strength

Notes: Each panel shows the first-stage Kleibergen-Paap F from 500 re-estimations of Table 2, column (2), with the national shifter replaced by a placebo series. Reordered series randomly permute the actual annual values across the 16 sample years; random walks are cumulative sums of standard normal draws. Solid line: actual encounter instrument ($F = 32.5$). Dashed line: actual budget instrument ($F = 14.8$). Short-dashed line: $F = 10$. Twenty-three draws with F below 0.01 are not shown.

E The Substitution Rate: Tail Diagnostics

The bootstrap distribution of the substitution rate is right-skewed, and its right tail has one identifiable source. All results use the benchmark estimate in Table 2, column (2), and the bootstrap of Section 4.6, unless noted.

E.1 Weak draws

Each bootstrap replication gives a different value of $\hat{\delta}$. The rate in each replication is $|\hat{\delta}|$ divided by $0.01 \times \bar{N}^*$, where $\bar{N}^*$ is held at its full-sample value of 15,426. Therefore, the only source of variation in the rate across replications is $\hat{\delta}$. Since $\hat{\delta}$ is the ratio of the reduced-form coefficient to the first-stage coefficient, replications with a weak first stage can produce large values of $|\hat{\delta}|$. Of the 999 replications, the median Kleibergen-Paap F statistic is 15.3, which is very similar to the 14.8 obtained using the entire sample.

However, 287 replications have an F statistic less than 10. Conditioning on the strength of the first stage removes much of the right tail while leaving the lower end of the distribution unchanged. The 95 percent percentile intervals are $[0.28, 3.19]$ for all replications, $[0.27, 2.30]$ for replications with $F > 5$, and $[0.27, 1.95]$ for replications with $F > 10$. At the same time, the share of draws below one increases from 42 percent to 53 percent.

E.2 Recomputing the denominator in each replication

In addition to calculating the confidence intervals using the full sample estimate of $\bar{N}^*$, I also run a second set of bootstrap replications in which each replication calculates its own weights and its own $\bar{N}^*$. The median rate is 1.32 in headcount and 0.93 in labor time; however, the tails are much larger than before. The 95 percent headcount interval is $[0.37, 23.39]$, and in 110 of 999 replications the sum of the weights is almost zero, making $\bar{N}^*$ undefined. The tail has the same source as before. When replications underweight the California core, they will have a weaker first stage. A weaker first stage will increase $|\hat{\delta}|$ and will also cause the weights to be unstable.

Therefore, as $|\hat{\delta}|$ increases, the numerator of the rate increases while the denominator of the rate decreases. There is a negative relationship between $|\hat{\delta}|$ and $\bar{N}^*$ across replications (correlation $= -0.50$). The smallest computed value of $\bar{N}^*$ is 22 workers. If I restrict the analysis to only those replications that have a first-stage F greater than 10, then the confidence interval is reduced to $[0.32, 6.90]$. The instability in the estimates is caused by the resampling process and not by the full sample estimate, since the full sample estimate of $\bar{N}^*$ is based on a strong first stage ($F = 14.8$).

E.3 No single zone drives the coefficient

The leave-one-zone-out jackknife estimates the benchmark and $\bar{N}^*$ 740 times, each time removing one zone. The median rate is 1.02, and 739 of the 740 rates lie within $[0.85, 1.86]$. The only rate outside this range occurs when the Salinas commuting zone is removed: $\bar{N}^*$ decreases from 15,426 to 2,182 workers, the first-stage F decreases to 7.7, and the rate increases to 9.6, the same weak-first-stage mechanism as in the bootstrap. Although the rate changes substantially, the coefficient itself does not change much: when $\bar{N}^*$ is held at its full-sample value, all 740 rates, including the one without Salinas, lie within $[0.84, 1.36]$. Therefore, the large change in the rate is due to the denominator and not to $\hat{\delta}$.

E.4 Which places

In each bootstrap replication, a zone may be selected zero, one, or more times. I calculate the correlation between the number of times a zone is selected and the rate in that replication. A negative correlation means that the rate is lower than average when the zone is selected more often than average, which indicates that the absence of this zone increases the rate. Conversely, a positive correlation means that the rate is higher than average when the zone is selected more often than average. The zones whose inclusion lowers the rate, and whose combined exclusion produces the tail, form the California core. The California core includes the Fresno-Tulare San Joaquin Valley zone (which alone accounts for 5.7 percent of national agricultural employment in 2000), greater Los Angeles, the Salinas Valley, Kern County, and Merced-Stanislaus. These zones are characterized by high baseline exposure, with PANCAL shares of 60 to 72 percent in 2000, large size, and low H-2A intensity (0.04 to 0.14 H-2A workers per agricultural worker, although the Salinas Valley has a higher intensity of 0.50). The zones whose inclusion raises the rate are those with high H-2A intensity. Examples include the Yakima Valley, the Columbia Basin, and Adams-Grant in Washington, as well as Palm Beach and Polk-Highlands in Florida. These zones have 0.8 to 2.2 H-2A workers per agricultural worker, an order of magnitude more than the California core.

The national rate is, in effect, an average across all of the commuting zones. However, there are large differences in how much H-2A workers are used in each zone. California has a large settled workforce, relatively low use of H-2A workers, and much of the weight, so it anchors the national rate close to one. Draws that underweight California reduce the variation available to identify the effect, weaken the first stage, and allow the H-2A-intensive regions of Washington and Florida to drive the results. A check on the full sample supports this. If I drop the three largest California commuting zones (Fresno-Tulare, greater Los Angeles, and the Salinas Valley), the coefficient in the overidentified specification remains

virtually identical (−182.7 against −178.9), but the identifying workforce drops from 14,913 to 3,953 workers and the first stage weakens to an F of 8.9, so no rate is reported. Therefore, the tail is due to the geography of the denominator, not to an unstable coefficient.

I also test whether the tail is driven by the largest H-2A cells. Dropping the top 1 percent of commuting-zone-year H-2A outcomes gives a coefficient of −168.2 (94.7), so the largest cells are not driving the estimate. Because this test is weakly identified (first-stage F of 4.1), it is read as a comparison of point estimates only.

F Measurement Diagnostics

The PANCAL measure excludes non-citizens who lived abroad in the prior year. Three diagnostics show that the measure does not count H-2A workers.

F.1 Group quarters

The ACS began including group quarters such as labor camps, worker dormitories and employer housing in 2006; therefore, guest workers would show up in this category if they were present in the survey. Figure G1 shows the percentage of agricultural workers who live in group quarters by group. Over the period of 2006 to 2024, the percentage of settled non-citizen agricultural workers who lived in group quarters was constant at approximately 5 percent. However, over the same time period, the percentage of recently arrived non-citizen agricultural workers who lived in group quarters averaged 34 percent and increased from 28 percent prior to 2018 to 46 percent after 2018 as H-2A employment increased. Only 1.9 percent of domestic workers lived in group quarters. This increase in group-quarters residence occurred only among recent arrivals and not among settled workers. Since the number of unweighted respondents in each cell of the recent arrival group is small (between 187 and 601 per year), the annual series is variable. Excluding recent arrivals therefore removes most H-2A workers who appear in the ACS from the settled measure.

F.2 Seasonality

The ACS considers a worker to be an agricultural laborer only if they are working in agriculture at the time of the interview; therefore, the ACS may underestimate the number of seasonal workers who are interviewed during their off-season. If settled non-citizen workers were either more or less seasonal than domestic workers, then this would cause a bias in the share.

According to monthly data from the Current Population Survey, the two groups have similar seasonal patterns. Both groups have their highest levels in July and their lowest levels

in March. The gap between the peak and trough for each group was 29.8 and 32.8 index points, respectively, over the period 2008–2024 (Figure G2).

G The Crop Controls

The vegetable share of cropland is used as the crop control in the main specification. Two figures illustrate why it is the composition of the crops grown by certified H-2A workers that is important, and what it removes. Figure G3 displays the crop composition of certified H-2A workers from 2011 to 2019, which are the years that the disclosure files list a crop. In 2011, the vegetable share was 16 percent. By 2016, it had increased to 32 percent and remained above a quarter through 2019. Conversely, the general/unspecified category decreased from 20 percent to less than 10 percent before increasing again. Berries increased from 3 to 9 percent, while tobacco, apples, and nursery stock remained constant.

Therefore, the expansion of the program is concentrated in vegetables, which is also the crop group in which settled non-citizen workers are most likely employed. If a commuting zone had vegetable acreage at baseline, then it would have experienced an increase in H-2A hiring over this time regardless of whether its settled workforce changed. The interaction of the vegetable share with year dummies accounts for this effect. Figure G4 presents the reduced-form event study results with and without the vegetable controls. The coefficients from 2017 onward are roughly double when the vegetable controls are removed. The coefficients prior to 2017 are nearly zero regardless of the controls.

Thus, the crop-demand confound inflates the magnitude of the response after 2017 but does not cause either the timing or pattern of responses before 2017.

Table G1 compares the vegetable share with two broader sets of crop controls, each fixed at its 2002 Census of Agriculture value and interacted with year dummies: a labor-intensive crop share, which is the combined acreage of vegetables, orchards, and berries as a share of cropland, and the vegetable, orchard, and tobacco shares entered as separate controls. Both broader sets absorb much of the identifying variation and weaken the first stage, with Kleibergen-Paap F-statistics below 10. The vegetable share captures the crop composition that both attracts settled workers and uses H-2A, and it keeps the first stage strong. Without any crop controls, the estimates are larger and less precise.

TABLE G1. Controls Progression

	Border Patrol budget		Both instruments	
Controls interacted with year	$\hat{\delta}$	KP F	$\hat{\delta}$	KP F
None	−414.8 (178.5)	11.64	−467.6 (178.7)	17.53
Labor-intensive crop share	−50.7 (96.1)	7.24	−60.3 (107.7)	4.53
Vegetable, orchard, and tobacco shares	−118.1 (84.1)	6.85	−136.7 (88.8)	4.53
Vegetable share (main specification)	−156.7 (61.3)	14.82	−178.9 (60.7)	16.31

Notes: Each row is the specification of Table 2 with the control set named in the first column, all fixed at their 2002 Census of Agriculture values and interacted with year dummies. The labor-intensive crop share is the combined acreage of vegetables, orchards, and berries as a share of cropland; the other rows use the single-crop shares of cropland. Hansen J p-values for the both-instruments column, top to bottom: 0.16, 0.50, 0.16, 0.17. Standard errors clustered by commuting zone in parentheses.

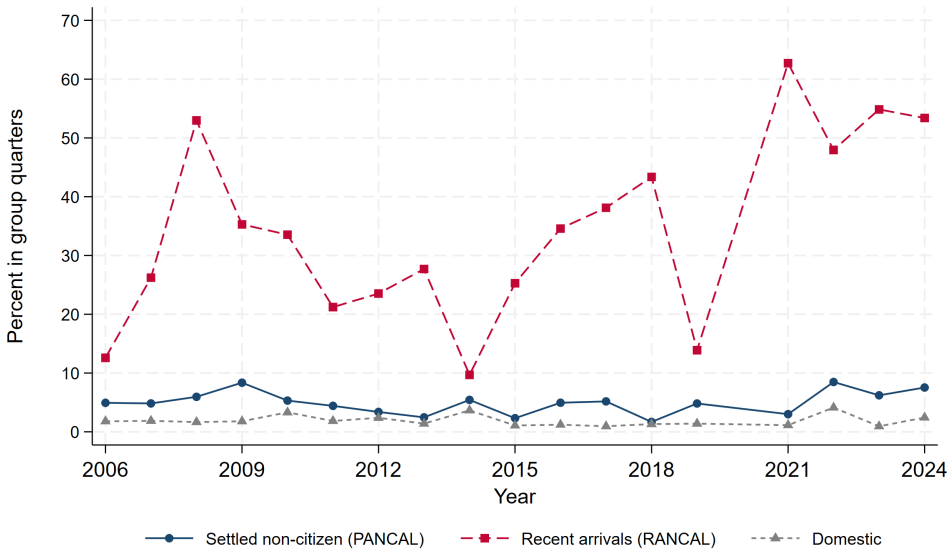


FIGURE G1. Share of Agricultural Laborers Living in Group Quarters, by Group

Notes: Person-weighted share of agricultural laborers in group quarters (IPUMS gq codes 3 and 4), by year and group. PANCAL is settled non-citizens; recent arrivals are non-citizens who lived abroad one year earlier; domestic is citizens. *Source:* ACS one-year files via IPUMS, 2006–2024.

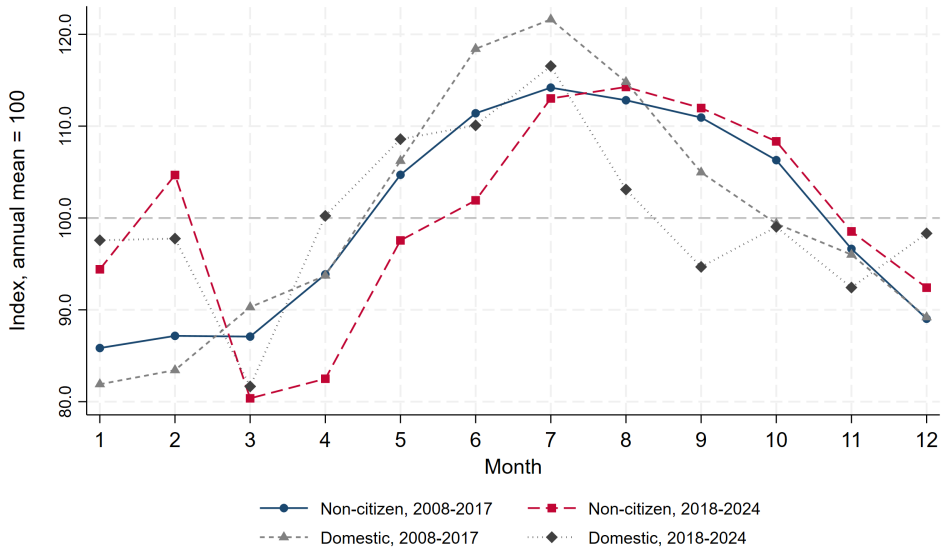


FIGURE G2. Seasonal Profile of Agricultural Laborers, CPS Monthly

Notes: Monthly employment of agricultural laborers indexed to the annual mean of 100, by citizenship and era. Same industry and occupation definitions as the ACS frame. Roughly 250 to 350 respondents per month. *Source:* CPS Basic Monthly files via IPUMS, 2008–2024.

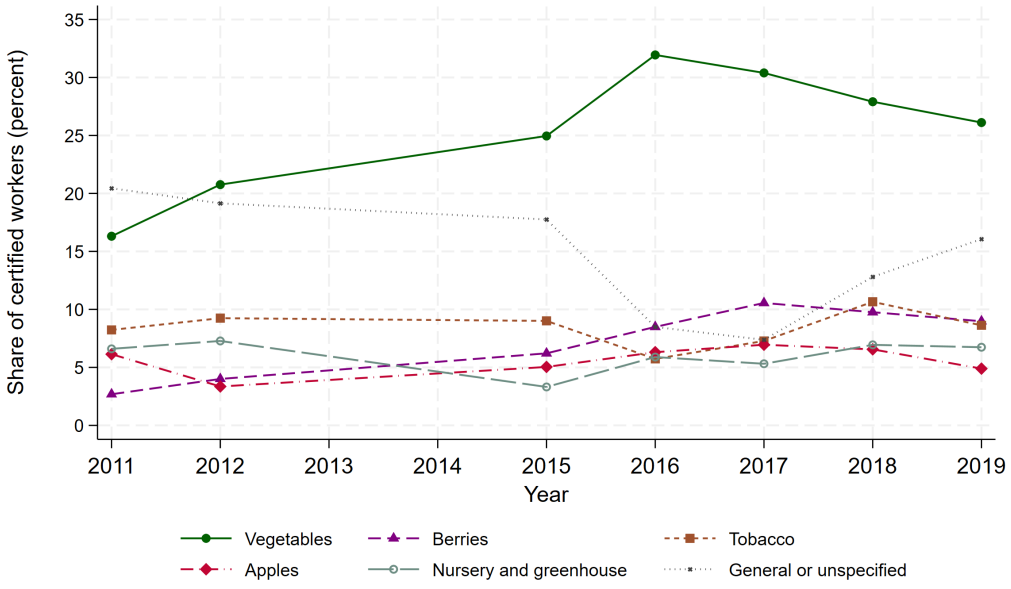


FIGURE G3. Crop Composition of Certified H-2A Workers

Notes: Worker-weighted shares of certified H-2A positions by crop, from the OFLC disclosure files. Crop information is unavailable for 2013 and 2014 and is not reported after 2019. Vegetables include melons, following the USDA-NASS classification. Only the six largest crop groups are shown; the general or unspecified category covers applications without a crop designation.

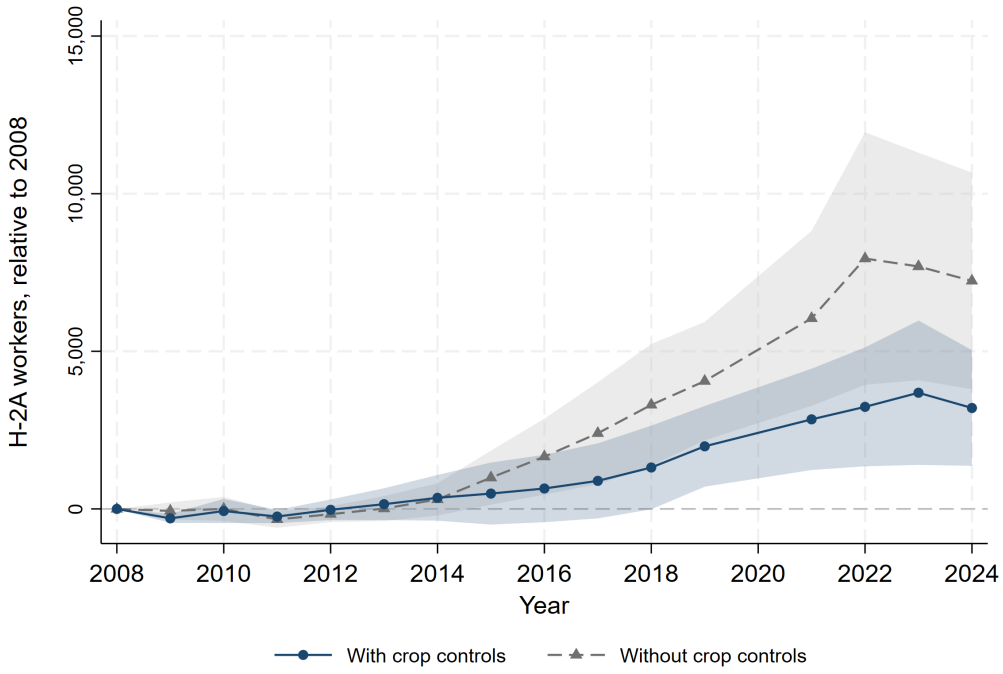


FIGURE G4. Reduced-Form Event Study With and Without Crop Controls

Notes: Coefficients on the year-2000 PANCAL share interacted with year dummies, with H-2A workers as the outcome and 2008 as the reference year, estimated with and without the vegetable share × year controls of the main specification. 740 commuting zones, weighted by year-2000 agricultural employment. Shaded bands are 95 percent confidence intervals clustered at the commuting zone level.